

Learning Heavy-Tailed Distributions with Diffusion Models



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To the people whose encouragement forms the foundation of these pages.

Declaration

I, Stefano Mauloni of St Edmund's College, being a candidate for the MPhil in Machine Learning and Machine Intelligence, hereby declare that this report and the work described in it are my own work, unaided except as may be specified below, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose.

Software Declaration The codebase for the main framework is my own, implemented using standard Python packages. This can be found [here](#). Third-party code used for verification only: The reference implementation released by [Shariatian et al. \(2025\)](#) was run unmodified on shared inputs, to confirm that DLPM's performance in Chapter 6 is a property of the algorithm. Its output enters the dissertation only through that check, reported in Section 6.3.

Word Count: 15000

Statement of Generative AI Usage

Describe your use (type, extent) of generative AI in the production of this dissertation. Please include as much detail as possible, including the platform you used. You may wish to refer to the list of use-cases included in the Use of Generative AI Policy.	Claude Code was used to assist with implementing code that is relatively simple to check for correctness (e.g. plotting scripts and small helper functions) and debugging syntax and runtime errors. Claude was also used for generating some of the figures in the dissertation under my guidance, and to convert prose passages into tables. Grammarly was used for proofreading purposes to identify spelling and grammatical errors in the written report.
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For each type of use, explain why you decided to use AI . If you did not use GenAI, you can, if you wish, also use this space to explain why you did not.	Claude Code was used to improve coding efficiency when implementing visualisations and resolving implementation errors, allowing more time to focus on experimental design, analysis, and interpretation. Grammarly was used to reduce superficial writing errors and improve clarity. Claude helped speed up time-consuming LaTeX-related tasks.
How did you validate any AI generated output (for example to assess for accuracy and bias)?	All AI-assisted code suggestions were manually reviewed, tested, and modified where necessary. Results were validated by reproducing runs, checking consistency across different model configurations, and ensuring that observed behaviours aligned with theoretical expectations. Any text flagged by Grammarly was reviewed manually to ensure that corrections did not alter technical meaning.
What were the strengths and weaknesses associated with AI use in each case? (E.g., consider whether the AI seemed to ‘misunderstand’ aspects of the topic, or identified points you might have overlooked; you may also wish to consider what impact the use of AI had on your learning, whether the AI ‘hallucinated’, etc.)	The primary strength of AI assistance was improved efficiency in routine tasks such as debugging, plotting, and proofreading. This helped streamline research. A limitation was that code suggestions often required correction or adaptation to fit the specific experimental context, reinforcing the need for human oversight. AI tools did not contribute to theoretical reasoning, experimental conclusions, or interpretation of results, which instead were carried out independently. Nonetheless, they were used as planning tools.

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Abstract

Heavy-tailed data is the norm in fields such as weather, hydrology and finance, and the extremes are usually the reason for building a generative model at all. Diffusion models do not reproduce them: the forward process adds Gaussian noise, and what is generated inherits its light tail. The established remedies replace the injected noise or the resulting stochastic process with heavy-tailed alternatives, and in each the tail weight of the generated law is a property of the corruption mechanism, set by a parameter. Where that parameter is instead fitted, as in tail-adaptive normalising flows, it is fitted by estimating a tail index from the few observations the tail contains.

This dissertation develops the *posterior lift*, in which the tails are carried by a latent variable that is never learned. Each datum is paired with a latent scale drawn from the posterior of a tractable heavy-tailed working model. Because the pairing is a posterior, the data distribution is left exactly unchanged whatever working model is chosen. That posterior is conjugate, so it is available in closed form, and nothing is estimated. Conditionally on the latent, every law the network encounters is sub-Gaussian, so classical Gaussian diffusion applies under a single substitution, and the tail index of the generated law is shown to be inherited from the data, with no tail index estimated at any point.

Since moment-based comparisons do not converge at these tail indices, the experiments build a quantile-based protocol and check each forward process against a closed form before any sample is compared. The lift wins every rung of a misspecification ladder on synthetic fields, including rungs on which its working model admits no latent scale at all, and leads the best baseline by a factor of 2.44 on the tail composite for real vertical-velocity data. It pays where a single latent scales an entire field, overstating the spatial coherence of extremes, and in low dimension, where the conditional scale has no variance.

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Notation

Symbols are grouped by role. Throughout, v is the latent scale, t the diffusion time and α the tail index of the data.

Symbol	Meaning	Defined at
<i>Data and tail weight</i>		
p_{data}, x_0	data density, a datum	
α	tail index of p_{data}	Definition 1
$\text{RV}_{-\alpha}, L$	regular variation, slowly varying factor	Definition 1
$x = \sqrt{v} \varepsilon$	Gaussian scale mixture	(2.3.1.1)
<i>Working model and posterior lift</i>		
(μ, β, p)	working model	Definition 9
v	working degrees of freedom	Proposition 12
$\text{GIG}(\lambda, \chi, \psi), K_\lambda$	mixing family, Bessel function	Section A.2.1
$v > 0$	latent scale	Definition 10
$m_v = \mu + \beta v$	conditional location	Definition 9
p_{wm}	working-model marginal	Definition 9
$q(v \mid x_0)$	lift kernel	(4.1.1.2)
λ', χ', ψ'	lift-kernel parameters	(4.1.2.1)
$\tilde{p}(x_0, v), \tilde{p}(v), \tilde{p}(x_0 \mid v)$	lifted joint, latent marginal, conditional	(4.1.1.3)
$\Lambda = p_{\text{data}}/p_{\text{wm}}$	likelihood ratio	Assumption (13)
$\hat{p}$	generation-time latent law	Theorem 16
κ	tail index of $\hat{p}$	Theorem 16
u_0, ξ	threshold and shape of the Pareto fit	Section 4.4
<i>Diffusion</i>		
$t \in [0, T], f(t), g(t)$	time, drift and diffusion coefficients	(2.1.1.1)

Symbol	Meaning	Defined at
$s(t), \sigma(t)$	signal and noise scale	(2.1.1.3)
$p_{0t}(x_t x_0, v)$	conditional perturbation kernel	Lemma 20
$\tilde{p}_t(\cdot v), \pi_v$	conditional marginal, terminal prior	(4.2.1.3)
$\text{SNR}(t v)$	conditional signal-to-noise ratio	Proposition 21
$y_t, q_t(\cdot v)$	standardised state and its marginals	Proposition 29
$s_\theta, \varepsilon_\theta, u_\theta$	score, noise and standardised networks	(4.3.2.4)
$\omega(t, v)$	score-matching weighting	Definition 25
$\mathcal{L}_{\text{SM}}, \mathcal{L}_{\text{DSM}}, \mathcal{L}(\theta)$	the three risks	(2.1.3.1), (4.3.1.6)
$\hat{p}_0$	generated law	Corollary 42
<i>Stochastic interpolants</i>		
ρ_0, ρ_1, ρ_t	endpoints and time-marginals	Definition 2
$b^*(t, x)$	probability-flow velocity	(3.1.0.2)
$m(t, x), \mu_t$	conditional median, ghost law	(3.3.0.1)
<i>Comparison with noise-replacement models</i>		
$u, c_t(u)$	generic mixing variable, noise scale	(5.1.0.1)
$\zeta(x_0)$	conditional noise scale	Definition 31
$w, A_t, T_{[0,1]}$	mixing objects of t-EDM, DLPM, HTFM	Section 5.1
<i>Evaluation</i>		
$r, k, \hat{\alpha}_k$	radius, order statistics used, Hill index	(6.2.0.1)
z	Hill discrepancy in standard errors	(6.2.0.2)
$\text{MSLE}, F^{-1}, \hat{F}^{-1}$	log-quantile error, quantile functions	(6.2.0.3)
$\chi(h), \ell_v$	extremal dependence, scale correlation length	Table 6.1
$f_m, M_m, C_{\mathcal{S}}$	floor, model value, composite	(6.2.0.4)

Chapter 1

Introduction

Rainfall, wind speed, insurance losses and financial returns share a property that generative models rarely reproduce. Their extremes follow a power law, so the largest observations sit orders of magnitude beyond the bulk and the variance need not exist at all. Those extremes are usually the reason for building the model: a system that reproduces the bulk faithfully and truncates the tail is confidently wrong exactly where a decision depends on it.

Diffusion models truncate. The forward process adds Gaussian noise, the reverse process is driven by a learned score, and what is generated inherits the light tail of the corruption. The established response is to replace the noise, with Student- t increments (Pandey et al., 2024), α -stable ones (Shariatian et al., 2025; Yoon et al., 2023), or a random time change (Yang et al., 2026). In each, the tail weight of what is generated is a property of the mechanism rather than of the data, fixed by a parameter chosen in advance, so matching a target's tail becomes a tuning problem whose answer must already be known. The normalising-flow literature goes one step further and fits that parameter, estimating a tail index from the data and configuring a heavy-tailed base with it (Jaini et al., 2020; Laszkiewicz et al., 2022); the difficulty then moves into the estimator, which is working on the few observations the tail contains.

Can a generative model take its tail behaviour from the data rather than from a hyperparameter?

The construction developed here answers yes, and does so by declining to learn the tails at all. A heavy-tailed law can be written as a Gaussian whose scale is random, $x = \sqrt{v}\epsilon$ (Andrews and Mallows, 1974; Barndorff-Nielsen, 1997). If each datum could be paired with the scale that produced it, the modelling problem would split: a light-tailed conditional, which diffusion handles, and a heavy-tailed law over scales, which requires no learning. Real data comes with no such pairing. What we do instead is posit a tractable heavy-tailed

working model and assign each datum the scale its posterior gives it. Because the assignment is a posterior, the data marginal is unchanged whatever working model is chosen; because the posterior is conjugate, the assignment is closed-form and nothing is estimated. Gaussian diffusion then runs on the conditionals, and the tail index of the generated law is inherited from the data rather than configured.

Contributions

Chapter 3 establishes why the alternative route fails. Transporting a light-tailed source onto a heavy-tailed target by a learned velocity field is ill-posed when the target has infinite variance, and the median, which survives where the mean does not, generates no continuity equation and transports the wrong law.

Chapter 4 constructs the posterior lift, shows that it leaves p_{data} exactly invariant and renders every conditional sub-Gaussian, and proves that the generated tail index is inherited from the data independently of the working degrees of freedom, in both the symmetric and the skewed case.

Chapter 5 places the construction among the heavy-tailed noise models by analysing what the mixing variable attaches to, whether it is marginalised or conditioned upon, and whether it is coupled to the datum. Only the last is new, and it is what makes the tail index a functional of the data.

Chapter 6 builds an evaluation protocol for targets whose moments do not converge, verifies each forward process against a closed form, and maps where the lift wins and where it pays.

On what is not new

Inferring a scale variable from the datum it describes is not new: Gaussian scale-mixture models of natural images have already done it (Wainwright and Simoncelli, 1999), as have independent component analysis and sparse coding (Hyvärinen and Oja, 2000; Olshausen and Field, 1996). There, the scale is a nuisance variable in the analysis of a fixed observation, and nothing is generated, so no question of tail behaviour arises.

Chapter 2

Background

This chapter presents the background the dissertation uses, and organises the heavy-tailed generative literature within a single framework that identifies a missing approach, which Chapter 4 then provides.

What do existing generative models of heavy-tailed data have in common, and what does that leave untried?

Section 2.1 fixes the two backbones and their notation, Section 2.2 the formalisation of tail weight, and Section 2.3 the scale-mixture primitive together with the closure properties that decide which constructions are available. Section 2.4 reads the existing heavy-tailed models through that lens, and Section 2.5 states the gap this dissertation addresses.

2.1 Diffusion and stochastic interpolants

Both backbones used below transport one distribution onto another and learn the object generating that transport by squared-error regression. We state the diffusion case in the form used in Chapter 4, then present stochastic interpolants as a generalisation.

2.1.1 The forward process

Corruption can be modelled as a linear stochastic differential equation on $\mathbb{R}^d$,

$$dx_t = f(t)x_t dt + g(t)dB_t, \quad x_0 \sim p_{\text{data}}, \quad t \in [0, T], \quad (2.1.1.1)$$

with drift coefficient f , diffusion coefficient g , and $(B_t)_{t \in [0, T]}$ a d -dimensional Brownian motion independent of x_0 (Song et al., 2021b). The increment dB_t is centred Gaussian with

covariance $dt I_d$. Because (2.1.1.1) is linear, its solution is Gaussian conditionally on x_0 and available in closed form,

$$x_t = s(t)x_0 + \sigma(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d), \quad (2.1.1.2)$$

with signal scale s and noise scale σ recovered from f and g by variation of constants,

$$s(t) := \exp\left(\int_0^t f(u) du\right), \quad \sigma(t)^2 := s(t)^2 \int_0^t \frac{g(u)^2}{s(u)^2} du. \quad (2.1.1.3)$$

We write $p_{0t}(x_t | x_0) = \mathcal{N}(x_t; s(t)x_0, \sigma(t)^2 I_d)$ for this *perturbation kernel* and p_t for the law of x_t . The schedule is chosen so that p_T is close to a Gaussian that can be sampled directly.

2.1.2 Time reversal and the score

Anderson (1982) showed that, under regularity conditions on the coefficients and on p_t which we assume throughout, (2.1.1.1) admits a time reversal. Running backwards from T to 0,

$$dx_t = [f(t)x_t - g(t)^2 \nabla_x \log p_t(x_t)] dt + g(t) d\bar{B}_t, \quad x_T \sim p_T, \quad (2.1.2.1)$$

with $\bar{B}$ a reverse-time Brownian motion, has the same marginals as (x_t) . So does the deterministic probability-flow ODE

$$\frac{dx_t}{dt} = f(t)x_t - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x_t), \quad (2.1.2.2)$$

which is what the deterministic DDIM sampler of Song et al. (2021a) integrates. Both are driven by the score $\nabla_x \log p_t$ of the noised marginals, so generation equates to estimating it.

2.1.3 Denoising score matching

Since p_{data} is unknown, so is the score, and it cannot be estimated by regressing directly against $\nabla_x \log p_t$. The standard route is denoising score matching: the conditional score of the perturbation kernel is available in closed form, and Vincent (2011) showed that regressing against it changes the risk only by a constant. For a positive weighting ω ,

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} \left[\left\| \omega(t) \left[s_\theta(x_t, t) - \nabla_{x_t} \log p_{0t}(x_t | x_0) \right] \right\|^2 \right], \quad \nabla_{x_t} \log p_{0t}(x_t | x_0) = -\frac{\varepsilon}{\sigma(t)}, \quad (2.1.3.1)$$

so $\mathcal{L}_{\text{DSM}}$ is minimised by the true score. Substituting the target shows that training is regression of the standard Gaussian ε that produced x_t , which is the noise-prediction form used in practice.

Equations (2.1.1.1)–(2.1.3.1) are used again in Chapter 4, where the same steps are carried out conditionally on a latent scale; we refer back to them there.

2.1.4 Stochastic interpolants

Read through (2.1.1.2), a diffusion model transports p_{data} onto a Gaussian, chosen only because it is easy to sample from. Nothing in the construction requires that particular endpoint, and one may ask whether p_{data} can be transported onto any tractable law instead. This is the stochastic-interpolant framework of Albergo and Vanden-Eijnden (2022); Albergo et al. (2025): given a coupling of endpoints $x_0 \sim \rho_0$ and $x_1 \sim \rho_1$, one interpolates them along a prescribed path and learns the velocity field that generates the resulting marginals. Flow matching (Lipman et al., 2022) and rectified flow (Liu et al., 2022) are the same construction under different choices of path and coupling.

The velocity is again a conditional expectation, this time of the instantaneous increment, and is again learned by L^2 regression. The target is now the difference of the endpoints, so it inherits their moments, and the theory is developed under a finite-fourth-moment assumption on both (Albergo and Vanden-Eijnden, 2022, Assumption 1). Chapter 3 asks what survives when that is weakened, the natural boundary for the targets of this dissertation.

The two backbones therefore share one structure: a conditional mean learned by squared-error regression against a target built from the data. That is what heavy tails disturb.

2.2 Heavy tails and regular variation

Tail weight is formalised throughout by regular variation: a tail that decays like a power law up to a slowly varying correction.

Definition 1 (Regular variation). A measurable function $L : (0, \infty) \rightarrow (0, \infty)$ is *slowly varying* at infinity if $\lim_{u \rightarrow \infty} L(cu)/L(u) = 1$ for every $c > 0$. A positive random variable X is *regularly varying with index* $\alpha > 0$, written $X \in \text{RV}_{-\alpha}$, if its tail satisfies

$$\mathbb{P}(X > u) = u^{-\alpha} L(u), \quad u \rightarrow \infty, \quad (2.2.0.1)$$

for some slowly varying L . The index α is the *tail index*: it governs which moments exist, since $\mathbb{E}[X^p] < \infty$ if $p < \alpha$ and $\mathbb{E}[X^p] = \infty$ if $p > \alpha$. For a random vector or a real random variable, we apply the definition to $\|X\|$ or $|X|$.

Moment conditions and tail indices are interchangeable, so a target with $\alpha < 2$ has no variance. Moreover, regular variation is stable under multiplication by an independent light-tailed factor, by Breiman’s lemma (Breiman, 1965; Resnick, 2007), which is what allows the tail of a scale mixture to be read off from the tail of its mixing variable, as Theorem 16 does.

A third fact bears on transport specifically. Jaini et al. (2020) show that a Lipschitz-continuous increasing map carries a light-tailed source to a light-tailed target, so that no such map can produce heavy tails from a Gaussian; a diffeomorphism achieving this must have unbounded derivative. The obstruction of Theorem 7 is the continuous-time counterpart of this statement, and the same paper introduced the standard response to it: replacing the source by a heavy-tailed one, which Section 2.4 takes up.

2.3 Gaussian scale mixtures

2.3.1 The primitive

The representation underlying every heavy-tailed model considered here is

$$x = \sqrt{v}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d) \perp\!\!\!\perp v > 0, \quad (2.3.1.1)$$

in which the direction is Gaussian and the tail behaviour is carried by the positive scalar v . Conditionally on v the law is $\mathcal{N}(0, vI_d)$; marginally it can be arbitrarily heavy, because rare large draws of v produce broad Gaussian components whose superposition has polynomial tails (Figure 2.1). Two instances recur. Taking $v \sim \text{InvGamma}(v/2, v/2)$ gives the multivariate Student- t with v degrees of freedom, the Andrews–Mallows representation (Andrews and Mallows, 1974); taking v a positive $\alpha/2$ -stable law gives the symmetric α -stable family by Bochner subordination (Samorodnitsky and Taqqu, 1994). Allowing a v -dependent mean, $x = \mu + \beta v + \sqrt{v}\varepsilon$, extends (2.3.1.1) to the normal mean–variance mixtures and hence to asymmetric laws; with generalised inverse Gaussian mixing this generates the generalised hyperbolic family (Barndorff-Nielsen, 1997; Jørgensen, 1982), which is the working model of Definition 9.

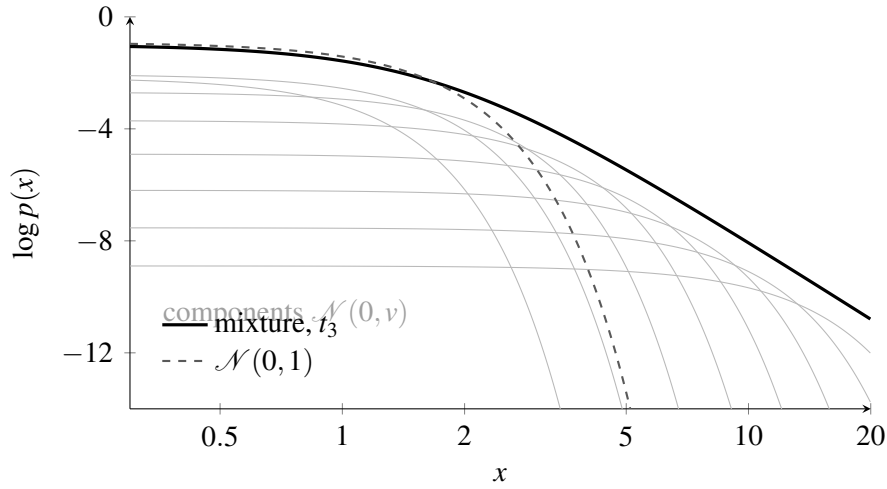


Figure 2.1 Why (2.3.1.1) produces heavy tails, on logarithmic axes. The thin curves are Gaussian densities $\mathcal{N}(0, v)$ at eight values of v between 0.5 and 64, each weighted by the inverse-gamma mixing law that yields the Student- t ; the bold curve is the mixture over all v and the dashed curve is a standard Gaussian. Every Gaussian, component or not, falls off a cliff. What makes the mixture heavy-tailed is that the cliffs recede as v grows, so that their upper envelope descends as a straight line, which on these axes is the signature of the regular variation of Definition 1. Every construction in Section 2.4 exploits this mechanism; they differ only in where the variable v is placed.

2.3.2 Closure properties

Which constructions are available depends on how each family behaves under the three operations a generative model performs on it: convolution, marginalisation, and conditioning.

The Gaussian family is closed under all three, which is why the classical theory is as uniform as it is. The α -stable family is closed under convolution at fixed α (this is the stability property from which the family takes its name) but its densities have no closed form outside a few special cases. On the other hand, the Student- t family is closed under marginalisation and conditioning, with a density available throughout, but it is *not* closed under convolution, being non-stable outside the Cauchy and Gaussian limits.

That asymmetry decides which backbones are available. A forward process can be specified either through the SDE (2.1.1.1), from which (2.1.1.2) follows by integration, or through a discrete transition kernel $q(x_t | x_{t-1})$ whose composition is read off as a perturbation kernel, as Ho et al. (2020) do. For Gaussian noise the two describe the same object. Once the noise is heavy-tailed it is not: composing increments requires convolution closure, specifying the perturbation kernel directly requires none, and so the same law can be admissible for one route and not the other. Chapter 5 develops this into the explanation of which existing models occupy which cells.

Finally, (2.3.1.1) suggests a strategy that recurs below in several forms: rather than confronting the heavy-tailed marginal, condition on v and work with the Gaussian conditional.

2.4 The heavy-tailed generative landscape

Method	Where the mixing variable sits	Coupled to x_0	Seen by net-work	Tail index set by
Gaussian diffusion	degenerate ($v \equiv 1$)	—	—	—
t-EDM, t-Flow (Pandey et al., 2024)	noise kernel, one draw per corruption	no	no (marginalised)	v , tuned
DLPM (Shariatian et al., 2025)	noise increment, one draw per step	no	no (marginalised)	α , tuned
LIM (Yoon et al., 2023)	not present; fractional score used	—	—	α , tuned
HTFM (Yang et al., 2026)	source covariance, path-valued clock	no	yes (clock feature)	clock law, tuned
t^3 -VAE (Kim et al., 2024)	joint prior over latent and output	yes	no (marginalised)	v , tuned
Posterior lift (Chapter 4)	data augmentation, one draw per datum	yes	yes (conditioning code)	inherited from data

Table 2.1 The heavy-tailed generative landscape read through (2.3.1.1). The rightmost column records the consequence: wherever the mixing variable is drawn independently of the data, the tail weight of the generated law is a property of the mechanism and is set by a hyperparameter, and only the last row makes it a functional of the data.

Table 2.1 organises existing methods in terms of four key design choices. Three groups appear.

Replacing the noise. The largest group substitutes a heavy-tailed law for the Gaussian in the corruption. Pandey et al. (2024) corrupt with Student- t noise, obtaining t-EDM and t-Flow according to the schedule used¹, and train against a γ -power divergence (Eguchi, 2021; Kim et al., 2024) because the squared loss would have an infinite-variance target. Shariatian et al. (2025) replace the increments of a DDPM chain by α -stable ones and train with a fractional L^r norm; their derivation conditions on the mixing variables to obtain a tractable denoising target, but the resulting model does not receive them. Yoon et al. (2023) work with the stable law directly through a fractional score, without the augmentation. In

¹there is no DDPM analogue, for a reason Chapter 5 makes precise

each case the mixing variable, where present, is drawn independently of the data, and the tail index of what is generated is a hyperparameter of the mechanism.

Conditioning on the mixing variable. Yang et al. (2026) take the strategy of Section 2.3 further, by replacing the deterministic time index of the source process with a random one. A *clock* is a nondecreasing random path used as a time change: running a Gaussian process along it rather than along t subordinates the process and yields a heavier-tailed law, in the same way that $\sqrt{v}\varepsilon$ in (2.3.1.1) is a Gaussian evaluated at a random scale. The clock is thus the mixing variable of (2.3.1.1), path-valued rather than scalar, and it controls the conditional covariance of the source. Conditionally on a realised clock path the source and the interpolating flow are Gaussian, so the ordinary conditional-flow-matching identity applies pathwise, while marginalising over clocks recovers Gaussian, α -stable and Student- t sources. The clock is encoded for the velocity field by a truncated logsignature where it is a genuine jump path, and by the scalar $\log V_v$ where the Student- t clock reduces it to one (Yang et al., 2026, App. E.4), so unlike DLPM the conditioning survives into the model. This covers $v \leq 2$, which the finite-variance Student- t preconditioning cannot, a statement only about the source law, not the data. The clock is nonetheless drawn independently of the data, and conditioning on a source-side variable leaves the data-moment assumption in place, so the construction addresses the tail of the source and not that of the target; Chapter 5 makes the distinction precise.

Data-dependent scale variables are long established outside generative diffusion, in independent component analysis, sparse coding and Gaussian scale-mixture models of natural images (Hyvärinen and Oja, 2000; Olshausen and Field, 1996; Wainwright and Simoncelli, 1999). The closest generative instance is t^3 -VAE (Kim et al., 2024), which couples a latent precision to the observation but marginalises it into a power-form joint fitted by the same γ -power divergence, so that the fitted object stays heavy-tailed and v stays a hyperparameter.

Approaches without a mixing variable. A parallel literature on normalising flows responds to Section 2.2 without any augmentation, by adopting a heavy-tailed base (Jaini et al., 2020) or with a final non-Lipschitz layer (Hickling and Prangle, 2025). It is also the one place where the tail index is not chosen in advance: tail-adaptive flows classify each marginal with extreme-value estimators and then fix or train the base degrees of freedom accordingly (Laszkiewicz et al., 2022), so the generated tailedness does follow the data. The route is estimator-mediated, and the estimate is formed from the tail observations; the lift reaches the same end without estimating a tail index at all, which is what Corollary 18 makes precise.

2.5 The gap

Read across Table 2.1, one column is constant. In every existing diffusion construction the mixing variable is drawn independently of the datum it accompanies. This is so even for the methods that condition on it. Observability and coupling are therefore separate properties, and only the first has been exploited.

The consequence is the rightmost column. If $v \perp\!\!\!\perp x_0$, the mixing law is a property of the mechanism rather than of the data, and the tail weight of what is generated is whatever the mechanism was configured to produce. Tail recovery becomes a tuning problem.

Coupling is the natural next step, and Yang et al. (2026) propose exactly it as future work, deferring it because inferring a learned, path-valued clock requires additional identifiability and estimation assumptions. That reservation does not apply to a scalar mixing variable inferred from a conjugate working model, where the conditional law is available in closed form; Section 5.4 returns to this.

Chapter 4 takes that path. Each datum is coupled to a latent scale drawn from the posterior of a heavy-tailed working model, the diffusion is run conditionally on that scale, and the tail index of the generated law follows from the data rather than from a hyperparameter. Chapter 3 comes first, and establishes why the alternative route, transporting a light-tailed source onto a heavy-tailed target, cannot be repaired.

Chapter 3

The impossibility of light-tailed transport for heavy tails

The reduction of Chapter 4 (i.e., routing heavy tails into a latent scale so that the learned problem is Gaussian) was not the first approach we tried. The natural starting point is transport: build a map from a simple prior to the heavy-tailed target and learn its velocity. This chapter shows why that approach cannot succeed for the targets of interest, and the failure motivates the solution. We work in the stochastic interpolant framework (Albergo and Vanden-Eijnden (2022)), whose definitions are recalled in Section 2.1.4. Here, we identify what specific issues arise when the fourth-moment assumptions of the theory are removed.

3.1 Transport under a finite first moment

Definition 2 (Linear stochastic interpolant). Given endpoints $x_0 \sim \rho_0$ and $x_1 \sim \rho_1$ on $\mathbb{R}^d$, drawn from a coupling with marginals ρ_0, ρ_1 , the deterministic *linear interpolant* is defined as

$$x_t = (1 - t)x_0 + tx_1, \quad t \in [0, 1], \quad (3.1.0.1)$$

The process (x_t) has time-marginals ρ_t interpolating ρ_0 and ρ_1 .

The transport is realised by a velocity field whose conditional expectation generates the marginals.

Definition 3 (Probability-flow field). The *probability-flow velocity* is the conditional mean

$$b^*(t, x) = \mathbb{E}[x_1 - x_0 \mid x_t = x], \quad (3.1.0.2)$$

and the associated *probability-flow ODE* is $\dot{y}_t = b^*(t, y_t)$, $y_0 \sim \rho_0$.

Proposition 4 (Transport identity). *If b^* in (3.1.0.2) is well defined and locally Lipschitz in x , uniformly in t , then the marginals ρ_t of (3.1.0.1) satisfy the continuity equation*

$$\partial_t \rho_t + \nabla \cdot (b^* \rho_t) = 0, \quad (3.1.0.3)$$

and the ODE flow of b^ pushes ρ_0 onto ρ_1 . The framework extends to a stochastic interpolant $\dot{y}_t = b^*(t, y_t) + (\text{score drift}) + \sqrt{2\varepsilon} \dot{W}_t$ solving the corresponding Fokker-Planck equation.*

We take (3.1.0.2) as the object to be learned and ask what it requires.

Assumption 5 (Finite first moment). The endpoints satisfy $\mathbb{E}\|x_0\| < \infty$ and $\mathbb{E}\|x_1\| < \infty$.

Assumption 5 is strictly weaker than the finite-fourth-moment condition (Albergo and Vanden-Eijnden, 2022, Assumption 1) and it is the natural boundary to probe. We show that even the objective used to learn b^* requires more than Assumption 5, and that b^* itself ceases to be a valid transport field for asymmetric heavy-tailed targets regardless of how it is estimated.

3.2 Two obstacles: the objective and the field

If the construction fails, does it fail because the target cannot be estimated, or because no field of the required form exists?

Theorem 6. *(The L^2 objective is ill-posed) The field b^* is the minimiser of the regression risk*

$$\mathcal{L}(b) = \mathbb{E}\|b(t, x_t) - (x_1 - x_0)\|^2. \quad (3.2.0.1)$$

If either endpoint has infinite second moment, then $\mathcal{L}(b) = \infty$ for every b with $\mathbb{E}\|b(t, x_t)\|^2 < \infty$, and the objective has no finite minimiser. This holds independently of any noise added to the interpolant, since the regression target $x_1 - x_0$ is intrinsic to (3.1.0.1).

The natural response is to weaken the loss. Under Assumption 5 the conditional median of $x_1 - x_0$ exists, and the mean absolute error $\mathbb{E}\|b(t, x_t) - (x_1 - x_0)\|_1$ is finite and is minimised by it. This repairs the objective but it changes what is learned, with consequences developed in Section 3.3. Before that, a second obstacle arises even when the objective is finite.

Theorem 7. *(Velocity blow-up) Let ρ_0 be light-tailed (e.g. Gaussian) and ρ_1 regularly varying (Definition 1) with index α . Then the transport field (3.1.0.2) grows without bound in x : there is no constant L with $\|b^*(t, x)\| \leq L(1 + \|x\|)$ uniformly in t . The local Lipschitz*

hypothesis of Proposition 4, and the regularity assumption of Albergo and Vanden-Eijnden (2022), fail.

Matching the prior's tail index to the target's mitigates the blow-up, since replacing the Gaussian prior by a Student- t of the same index keeps the ratio bounded. However, this does not resolve the next theoretical problem.

3.3 The median cannot transport

The conditional median survives where the mean does not. Does replacing one with the other rescue the construction?

Under Assumption 5 the only finite objective is the L^1 one, whose minimiser is the conditional median

$$m(t, x) = \text{med} [x_1 - x_0 \mid x_t = x]. \quad (3.3.0.1)$$

The transport equation (3.1.0.3), however, is generated by the conditional *mean* b^* , not the median. The two coincide only when the conditional law of $x_1 - x_0$ given x_t is symmetric.

Theorem 8. *(No transport equation for the conditional median) Let the conditional law of $x_1 - x_0$ given x_t be asymmetric on a set of x_t of positive ρ_t -measure, so that $m(t, \cdot) \neq b^*(t, \cdot)$. Then:*

1. *the flow $\dot{y}_t = m(t, y_t)$ does not solve the continuity equation (3.1.0.3), and its time-1 law is in general not ρ_1 ;*
2. *no continuity equation is generated by m : unlike the mean, the conditional median does not satisfy a tower property $\text{med}[\cdot] = \mathbb{E}[\text{med}[\cdot \mid \mathcal{G}]]$, so the marginal drift cannot be assembled from conditional medians, and there is no law transported by m that reduces to ρ_1 at $t = 1$.*

Integrating with the median solves the transport for a *ghost law* μ_t that has accumulated asymmetry bias at every instant. We attempted the standard escapes: log-space velocity, the Huber loss, a diffeomorphic reparameterisation, and a stochastic sampler. Each either reintroduces the divergence of Theorem 6 as it approaches the mean or inherits the median's bias as it approaches L^1 . The obstruction is that a single scalar field cannot be simultaneously finite-variance to learn and equal to the mean to transport, when the target is both heavy-tailed and asymmetric.

3.4 Limitations of the negative results

A boundary of Theorem 8 is worth stating precisely, as it motivates the approach taken in Chapter 4.

The result is about learning the transport *field*. It says nothing against methods that avoid estimating a conditional mean of a heavy-tailed quantity altogether. This is precisely the gap that Chapter 4 exploits: rather than transporting a light prior to a heavy target and paying the price of Theorems 7 and 8, we condition on a latent scale so that the quantity actually regressed is standard normal (Theorem 28), which is symmetric, finite-variance, and mean-median coincident by construction. The impossibility of light-tailed transport for heavy, asymmetric targets is thus not a dead end but a specification: make the conditional problem light-tailed, which is the subject of Chapter 4.

The boundary is also met by an existing method: t-Flow (Pandey et al., 2024) interpolates from a Student- t_ν prior to the data, the tail-matched symmetric configuration in which the quantile ratio governing Theorem 7 is bounded and mean and median coincide, so Theorem 8 does not bite.

Chapter 4

Lifted Diffusion

Chapter 3 showed that transporting a light-tailed prior onto a heavy-tailed target is doomed: the regression defining the transport field is either ill-posed or recovers the conditional median. The obstruction was never tail weight in isolation but the demand that one learned field be simultaneously finite-variance to estimate and equal to the conditional mean to transport.

This chapter takes the complementary route.

If the heavy tails cannot be learned, can they instead be carried by something that is never learned?

The answer developed here is a latent scale variable under which the data becomes conditionally light-tailed. Standard Gaussian diffusion runs on the conditionals, and the target is recovered by averaging over the latent. The heavy tails are carried by the latent, which is never score-matched. The construction proceeds in four steps, each occupying a section. We first define the coupling between data and latent scale and establish two properties: it leaves the data marginal exactly unchanged and renders the conditionals light-tailed (Section 4.1). We then run the classical diffusion machinery on the conditionals: the forward process and its time reversal (Section 4.2), and the denoising objective, which we show has standard-normal targets and standardised inputs (Section 4.3), and close with the concrete instantiations and algorithms (Section 4.4). Throughout, the guiding principle is that every object the network encounters is the one it would encounter in ordinary Gaussian diffusion; the difference from the standard theory is confined to how the latent scale is sampled and marginalised.

4.1 The posterior lift

Can each datum be assigned a latent scale that makes it conditionally light-tailed, without changing the target distribution?

4.1.1 Working model and lift

Section 2.3 recorded that a heavy-tailed law can be written as a Gaussian whose scale is random, $x = \sqrt{v}\varepsilon$. If the data admitted that representation exactly, the answer would be immediate: condition on v , and diffuse a Gaussian. It does not, so a rule is needed that attaches a scale to an arbitrary datum. Three objects supply one, and each answers a question raised by the previous.

The scale must come from somewhere, so we posit a tractable heavy-tailed law and read the scale off it. Extending the primitive (2.3.1.1) with a scale-dependent mean also supplies the asymmetry that real data exhibit.

Definition 9 (Normal mean–variance mixture working model). Given a location $\mu \in \mathbb{R}^d$, a skewness vector $\beta \in \mathbb{R}^d$ and a mixing law p on $\mathbb{R}_{>0}$, the *working model* is the law of

$$x = \mu + \beta v + \sqrt{v}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d) \perp\!\!\!\perp v \sim p(v). \quad (4.1.1.1)$$

We write $m_v := \mu + \beta v$ for the conditional location, so that $x | v \sim \mathcal{N}(m_v, vI_d)$, and $p_{\text{wm}}(x) := \int \mathcal{N}(x; m_v, vI_d) p(v) dv$ for the resulting marginal density.

The word *working* carries its statistical sense. This law is not asserted to be the data law, and Proposition 11 shows that nothing is lost when it is wrong.

The working model says how a datum arises from a scale. We need the converse: a rule attaching a scale to a datum already in hand. Bayes' rule supplies one, and it is the choice of the posterior that makes the construction exact.

Definition 10 (Posterior lift). Let p_{data} be the data density and let (μ, β, p) be a working model with marginal p_{wm} . The *lift kernel* is the working-model posterior

$$q(v | x_0) = \frac{\mathcal{N}(x_0; m_v, vI_d) p(v)}{p_{\text{wm}}(x_0)}, \quad (4.1.1.2)$$

and the *lifted joint*, *lifted latent marginal* and *lifted conditional* are

$$\tilde{p}(x_0, v) := p_{\text{data}}(x_0) q(v | x_0), \quad \tilde{p}(v) := \int \tilde{p}(x_0, v) dx_0, \quad \tilde{p}(x_0 | v) := \frac{\tilde{p}(x_0, v)}{\tilde{p}(v)}. \quad (4.1.1.3)$$

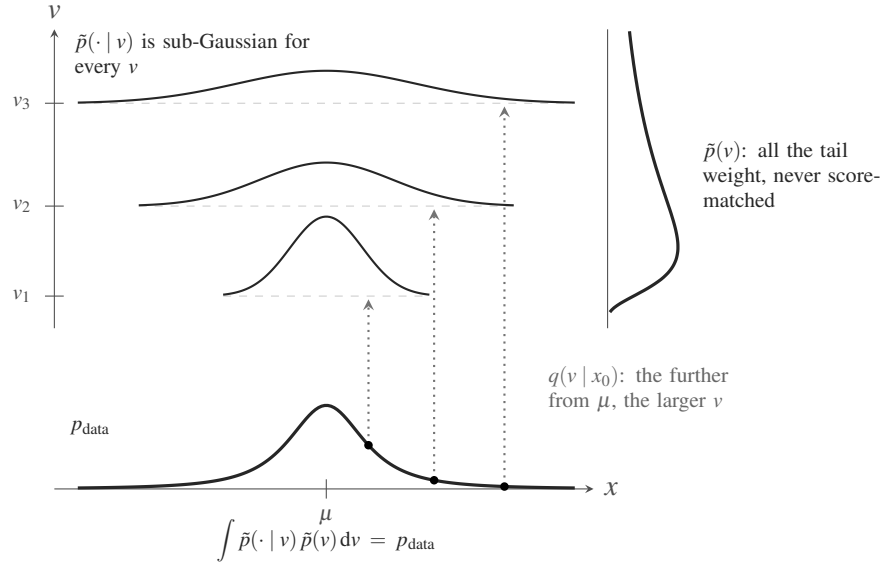


Figure 4.1 The posterior lift. Each datum is assigned a latent scale by the working-model posterior, so points far from μ receive large v . Read horizontally, the construction is a family of sub-Gaussian conditionals (Proposition 14), on which classical diffusion runs unchanged; read vertically, it is a heavy-tailed latent marginal, which the diffusion never touches. Mixing back over that marginal returns the data density exactly (Proposition 11).

A posterior is only useful if it can be evaluated, which is what fixes the last choice. Taking p in $\text{GIG}(\lambda, \chi, \psi)$ makes (4.1.1.2) available in closed form, so that no inference is performed and nothing is estimated; the density and its special cases are recorded in Section A.2.1, and the only facts used below are the parameter updates of (4.1.2.1). With this choice, (4.1.1.1) generates the generalised hyperbolic family, which contains the Student- t and skew- t , normal-inverse Gaussian, hyperbolic and variance-gamma laws.

Two questions follow at once. Does the lift leave the modelled distribution alone, and does conditioning on v actually remove the heavy tails? The next subsection answers both.

4.1.2 What the lift guarantees

Proposition 11 (Exactness of the lift). *For any working model (μ, β, p) with $p_{\text{wm}} > 0$ on the support of p_{data} , the x -marginal of the lifted joint is the data law: $\int \tilde{p}(x_0, v) dv = p_{\text{data}}(x_0)$. Consequently, learning the lifted conditionals $\tilde{p}(\cdot | v)$ and integrating v out targets p_{data} with no bias arising from the lift.*

Choosing the wrong μ, β , or mixing law changes how hard the conditional learning problem is, never which distribution the procedure targets. What the lift fixes is the law being approximated; how closely the generated samples approximate it still depends on the conditional

score error, through Corollary 42, and a working model far from the data makes that error larger. Section 6.5 measures the gap on skewed data, where a symmetric working model targets the right law but recovers its tail less accurately than a correctly skewed one. We exploit the exactness deliberately in Section 6.6.2.

Proposition 12 (Conjugacy of the GIG lift). *Let the working model of Definition 9 have mixing law $p = \text{GIG}(\lambda, \chi, \psi)$. Then the lift kernel (4.1.1.2) is again generalised inverse Gaussian,*

$$q(v | x_0) = \text{GIG}\left(\lambda - \frac{d}{2}, \chi + \|x_0 - \mu\|^2, \psi + \|\beta\|^2\right). \quad (4.1.2.1)$$

In particular the Student- t lift ($\lambda = -v/2$, $\chi = v$, $\psi = 0$, $\beta = 0$) is inverse-gamma throughout: $q(v | x_0) = \text{InvGamma}\left(\frac{v+d}{2}, \frac{v+\|x_0-\mu\|^2}{2}\right)$. For a product working model applied per coordinate, (4.1.2.1) holds coordinatewise with $\lambda - \frac{1}{2}$, $\chi + (x_{0,j} - \mu_j)^2$ and $\psi + \beta_j^2$.

Assumption 13 (Polynomial tail domination). *There exist constants $C > 0$ and $k \geq 0$ such that the likelihood ratio $\Lambda := p_{\text{data}}/p_{\text{wm}}$ of the lift satisfies $\Lambda(x) \leq C(1 + \|x\|)^k$ for all $x \in \mathbb{R}^d$; that is, the data tails are no heavier than those of the working model up to polynomial factors.*

Proposition 14 (Conditional light tails). *The lifted conditional factorises as*

$$\tilde{p}(x_0 | v) \propto \mathcal{N}(x_0; m_v, vI_d) \Lambda(x_0), \quad \Lambda = p_{\text{data}}/p_{\text{wm}}. \quad (4.1.2.2)$$

Under Assumption 13, for every $v > 0$ there is a constant $C(v) < \infty$ with $\tilde{p}(x_0 | v) \leq C(v) \exp\{-\|x_0 - m_v\|^2/(4v)\}$; in particular $\tilde{p}(\cdot | v)$ is sub-Gaussian and has finite moments of all orders, however heavy the tails of p_{data} may be.

Conditioning therefore destroys the heavy tails as a consequence of lifting through a heavy-tailed working model. This is the property that makes every subsequent object the classical Gaussian one.

Remark 15 (The lifted latent marginal is not the working prior). Combining (4.1.1.2) and (4.1.1.3),

$$\tilde{p}(v) = p(v) \mathbb{E}_{x_0 \sim p_{\text{data}}} \left[\frac{\mathcal{N}(x_0; m_v, vI_d)}{p_{\text{wm}}(x_0)} \right], \quad (4.1.2.3)$$

which coincides with the working prior $p(v)$ if and only if $p_{\text{data}} = p_{\text{wm}}$. When the data are heavier-tailed than the working model, $\tilde{p}$ is substantially heavier than p : large $\|x_0\|$ forces large v through (4.1.2.1). Generation must therefore draw v from $\tilde{p}$, not from p ; Theorem 16 shows that this is precisely what determines the tail of the generated law.

4.1.3 Inheritance of the tail index

Theorem 16 (Tail inheritance). *Let $\hat{p}$ be the law from which the latent scale is drawn at generation time, and suppose the generated sample admits the representation $x = \mu + \beta v + \sqrt{v}y$ with $v \sim \hat{p}$ and, conditionally on v , $\mathbb{E}[\|y\|^{2\kappa+\delta} \mid v] \leq M < \infty$ for some $\delta > 0$ (guaranteed by Proposition 14). Suppose $\hat{p}$ is regularly varying (Definition 1) at infinity with index $-\kappa$, $\kappa > 0$. Then the tail of the generated law is determined by whichever of the two v -dependent terms dominates.*

1. Symmetric case, $\beta = 0$. The scale term governs, and

$$\mathbb{P}(\|x\| > u) \sim \mathbb{E}[\|y\|^{2\kappa}] \mathbb{P}(\sqrt{v} > u), \quad u \rightarrow \infty, \quad (4.1.3.1)$$

so that $\|x\|$ is regularly varying with tail index 2κ .

2. Skewed case, $\beta \neq 0$. The location term governs, and

$$\mathbb{P}(\|x\| > u) \sim \mathbb{P}(\|\beta\|v > u) = \|\beta\|^\kappa \mathbb{P}(v > u) (1 + o(1)), \quad u \rightarrow \infty, \quad (4.1.3.2)$$

so that $\|x\|$ is regularly varying with tail index κ .

In both cases the tail index of the generated law is a functional of $\hat{p}$ alone, but the functional differs: it is 2κ under a symmetric working model and κ under a skewed one.

Remark 17 (The moment hypothesis is a property of the model, not of the target). Theorem 16 requires $\mathbb{E}[\|y\|^{2\kappa+\delta} \mid v] \leq M < \infty$, which Proposition 14 supplies for the *exact* conditional: under Assumption 13 the standardised conditional is sub-Gaussian uniformly in v , so $\|y\|$ has no power-law tail. For a *learned* conditional, the hypothesis is an assumption about the trained network, and nothing in the framework enforces it. If the learned $\|y\|$ carries a power-law tail of index γ , then $x = \mu + \beta v + \sqrt{v}y$ is a product of two regularly varying factors and the generated index is $\min(2\kappa, \gamma)$ in the symmetric case. Therefore, a shortfall in the generated tail index is diagnostic of the conditional rather than of the lift, and the two can be separated by measuring the index of the latent pool and that of the standardised residual $\|x - \mu\|/\sqrt{v}$ directly. We do so in Section 6.5.

Corollary 18 (The lift is tail-adaptive). *Suppose p_{data} has regularly varying tails of index $\alpha > 0$ and let the working model have $\psi = 0$ and mixing $\text{GIG}(-v/2, v, 0) = \text{InvGamma}(v/2, v/2)$, for any working $v > 0$. Generation with $\hat{p} = \tilde{p}$ produces samples with tail index α in both of the following cases, independently of the working v .*

1. Student- t lift, $\beta = 0$. The posterior scale parameter is $\chi' = v + \|x_0 - \mu\|^2$ and $\psi' = 0$, so $\mathbb{E}[v \mid x_0] = \chi'/(v + d - 2) \asymp \|x_0 - \mu\|^2$. Provided $\alpha < v + d$, the normalised posterior has a moment of order $\alpha/2 + \delta'$ and $\tilde{p} \in \text{RV}_{-\alpha/2}$, so Theorem 16(i) with $\kappa = \alpha/2$ gives index $2\kappa = \alpha$. The condition is not restrictive, as it fails only for a working model heavier-tailed than the data in low dimension.
2. Skew- t lift, $\beta \neq 0$. The posterior parameters are $\chi' = v + \|x_0 - \mu\|^2$ and $\psi' = \|\beta\|^2 > 0$, so as $\chi'\psi' \rightarrow \infty$ the GIG mean satisfies $\mathbb{E}[v \mid x_0] \sim \sqrt{\chi'/\psi'} \asymp \|x_0 - \mu\|/\|\beta\|$. Hence $\tilde{p} \in \text{RV}_{-\alpha}$, and Theorem 16(ii) with $\kappa = \alpha$ gives index $\kappa = \alpha$.

The two cases cancel: a symmetric working model concentrates the posterior quadratically in the radius and enters x through $\sqrt{v}$, while a skewed one concentrates it linearly and enters through βv . Either way, the generated tail index is α .

Corollary 18 isolates what distinguishes the present framework from heavy-tailed-noise diffusions: there, the tail index of the generated law is a hyperparameter of the corruption kernel, whereas here it is inherited from the data through the lifted latent marginal. We verify the independence from v empirically in Section 6.5.

4.2 Diffusion on the lifted conditionals

Once the conditionals are light-tailed, does anything about standard diffusion have to change?

This section records what that machinery becomes, and nothing in it is new: each object is the corresponding object of Section 2.1 under the single substitution $g \mapsto \sqrt{v}g$. The one structural point is that all dynamics act on x alone, so (x_t, v) is a Markov process in which v has no dynamics, and the heavy tails sit entirely in $\tilde{p}(v)$, which the dynamics never touch and the network never score-matches.

4.2.1 The conditional forward process

Definition 19 (Conditional forward process). For each $v > 0$ the *conditional forward process* is (2.1.1.1) started from the lifted conditional and driven at scale $\sqrt{v}$,

$$dx_t = f(t)x_t dt + \sqrt{v}g(t)dB_t, \quad x_0 \sim \tilde{p}(\cdot \mid v), \quad (4.2.1.1)$$

with f, g continuous, $(B_t)_{t \in [0, T]}$ a d -dimensional Brownian motion independent of (x_0, v) , and the schedules s, σ as in (2.1.1.3).

Lemma 20 (Conditional perturbation kernel). *Since f is unchanged and g is scaled by $\sqrt{v}$, (2.1.1.3) leaves s untouched and multiplies the noise scale by $\sqrt{v}$. Equation (2.1.1.2) therefore reads*

$$x_t = s(t)x_0 + \sigma(t)\sqrt{v}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d) \perp\!\!\!\perp (x_0, v), \quad (4.2.1.2)$$

the conditional perturbation kernel is $p_{0t}(x_t | x_0, v) = \mathcal{N}(x_t; s(t)x_0, v\sigma(t)^2 I_d)$, and the conditional marginals converge to the scaled Gaussian prior

$$\tilde{p}_T(\cdot | v) \approx \pi_v := \mathcal{N}(s(T)m_v, v\sigma(T)^2 I_d). \quad (4.2.1.3)$$

Proposition 21 (Scale invariance of the conditional SNR). *Define the conditional signal-to-noise ratio of (4.2.1.2) as $\text{SNR}(t | v) := s(t)^2 \text{trCov}(x_0 | v) / (d v \sigma(t)^2)$. Under the working model, $\text{Cov}(x_0 | v) = vI_d$ and hence*

$$\text{SNR}(t | v) = \frac{s(t)^2}{\sigma(t)^2}, \quad (4.2.1.4)$$

which is independent of v . In particular, relative to the conditional scale, the noise level reached at any given t is the same for every latent draw. The computation is under the working model, which is what the $\sqrt{v}$ scaling is calibrated to; away from the exact-NMVM point of Corollary 30 the lifted conditional has covariance vI_d only up to the tilt Λ , and the v -freeness is correspondingly approximate.

Proposition 21 is the reason for the $\sqrt{v}$ factor in (4.2.1.1). Without it, a single time grid would correspond to widely different effective noise levels across the range of v , which would be severe for heavy-tailed lifts, and one network could not serve all conditionals at once. With it, the diffusion is the *same* process for every v up to an affine change of variables. This is made precise in Proposition 29.

Remark 22 (A latent-dependent prior costs nothing). The terminal law (4.2.1.3) depends on v , which is immaterial at generation time as v is drawn first and held fixed, so π_v is known before the reverse dynamics are initialised (Algorithm 2). By Proposition 21 the quality of the Gaussian approximation in (4.2.1.3) is governed by the v -free quantity $\sigma(T)^2$, so a single schedule choice controls the prior mismatch uniformly over the latent scale.

4.2.2 Conditional time reversal

Assumption 23 (Regularity for time reversal). For each $\nu > 0$ and $t \in (0, T]$ the conditional marginal $\tilde{p}_t(\cdot | \nu)$ admits a smooth, strictly positive density, and $\int_0^T \mathbb{E}_{\tilde{p}_t(\cdot | \nu)} \|\nabla \log \tilde{p}_t(x_t | \nu)\|^2 dt < \infty$.

The integrability requirement in Assumption 23 is not an additional restriction on the data: by Proposition 14 each $\tilde{p}(\cdot | \nu)$ is sub-Gaussian, and Gaussian smoothing through Lemma 20 preserves this, so the finite-moment conditions of the classical theory hold conditionally on ν even when the data marginal violates them. Only smoothness and positivity are genuinely assumed.

Theorem 24 (Conditional time reversal). *Under Assumption 23, (2.1.2.1) holds for each fixed ν : the process solving, backwards from $t = T$ to $t = 0$,*

$$dx_t = \left[f(t)x_t - \nu g(t)^2 \nabla_x \log \tilde{p}_t(x_t | \nu) \right] dt + \sqrt{\nu} g(t) d\bar{B}_t, \quad x_T \sim \tilde{p}_T(\cdot | \nu), \quad (4.2.2.1)$$

has law $\tilde{p}_t(\cdot | \nu)$ at every t , as does the probability-flow ODE

$$\frac{dx_t}{dt} = f(t)x_t - \frac{\nu g(t)^2}{2} \nabla_x \log \tilde{p}_t(x_t | \nu). \quad (4.2.2.2)$$

Conditioning is legitimate because ν is static: the generator of (x_t, ν) acts on the x -coordinate alone, so the reversal of the pair is the ν -wise reversal of x .

The only trace of the lift in (4.2.2.1) and (4.2.2.2) is a fixed multiplicative ν , so every guarantee available for Gaussian diffusion holds conditionally, and Section 4.3.3 converts these into a bound on the marginal error. At generation time, the reverse dynamics are initialised at π_ν rather than at $\tilde{p}_T(\cdot | \nu)$, and that discrepancy is the prior-mismatch term of Theorem 41.

4.3 The learning objective and its analysis

What does the network actually regress against, and is that quantity well behaved however heavy the data?

4.3.1 Conditional denoising score matching

Definition 25 (Conditional score-matching risks). For a weighting $\omega(t, v) > 0$ and a network $s_\theta(x_t, t, v)$, the risks of (2.1.3.1) formed conditionally on the latent are

$$\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E}_{t, v \sim \tilde{p}(v), x_t \sim \tilde{p}_t(\cdot | v)} \left[\omega(t, v) \|s_\theta(x_t, t, v) - \nabla_{x_t} \log \tilde{p}_t(x_t | v)\|^2 \right], \quad (4.3.1.1)$$

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{t, (x_0, v) \sim \tilde{p}, \varepsilon \sim \mathcal{N}(0, I_d)} \left[\omega(t, v) \|s_\theta(x_t, t, v) - \nabla_{x_t} \log p_{0t}(x_t | x_0, v)\|^2 \right], \quad (4.3.1.2)$$

with x_t as in (4.2.1.2).

Theorem 26 (Conditional DSM equivalence). *Under Assumption 23, for every weighting ω we have $\mathcal{L}_{\text{DSM}}(\theta) = \mathcal{L}_{\text{SM}}(\theta) + C$ with $C \geq 0$ independent of θ , and both risks are minimised by the true conditional score $s^*(x, t, v) = \nabla_x \log \tilde{p}_t(x | v)$. This is the identity behind (2.1.3.1) carried out at fixed v , which is legitimate because v is not integrated over inside the conditional expectation.*

Everything to this point has been the classical theory read at fixed v . The first place the lift changes the problem rather than restating it is the regression target.

Proposition 27 (Regression target and its second moment). *The denoising target of (4.3.1.2) is*

$$\nabla_{x_t} \log p_{0t}(x_t | x_0, v) = -\frac{x_t - s(t)x_0}{v\sigma(t)^2} = -\frac{\varepsilon}{\sqrt{v}\sigma(t)}, \quad (4.3.1.3)$$

and its conditional second moment is

$$\mathbb{E} \left[\|\nabla_{x_t} \log p_{0t}(x_t | x_0, v)\|^2 \mid v \right] = \frac{d}{v\sigma(t)^2}. \quad (4.3.1.4)$$

Equation (4.3.1.4) is the central contrast with unlifted training on heavy-tailed data: *the target moment does not depend on the data distribution at all*. The tails have been routed entirely into $\tilde{p}(v)$, and the only possible sources of divergence are the lower tail of the latent and the classical $t \rightarrow 0$ denoising singularity—neither of which has anything to do with how heavy p_{data} is.

Under a weighting that ignores the latent, the risk is finite only if $\mathbb{E}_{\tilde{p}(v)}[v^{-1}] < \infty$, which the lift satisfies whenever p_{data} has no atom at μ (Corollary 38, Remark 39). The weighting adopted next removes the condition altogether, so the restriction is recorded rather than relied upon.

Theorem 28 (Scale-adapted weighting and noise prediction). *Adopt the weighting $\omega(t, v) = v\sigma(t)^2$ and reparameterise*

$$\varepsilon_\theta(x, t, v) := -\sqrt{v}\sigma(t)s_\theta(x, t, v). \quad (4.3.1.5)$$

Then the risk (4.3.1.2) equals the noise-prediction objective

$$\mathcal{L}(\theta) = \mathbb{E}_{t, (x_0, v) \sim \tilde{p}, \varepsilon} \left[\left\| \varepsilon_\theta(s(t)x_0 + \sigma(t)\sqrt{v}\varepsilon, t, v) - \varepsilon \right\|^2 \right], \quad (4.3.1.6)$$

whose regression target is standard normal. Consequently (4.3.1.6) has finite value and finite-variance targets for any mixing law, with no moment condition on $\tilde{p}(v)$; its optimal value $\mathbb{E}\|\varepsilon - \mathbb{E}[\varepsilon \mid x_t, v]\|^2 \leq d$ is uniformly bounded; and its minimiser is the conditional Tweedie denoiser

$$\varepsilon^*(x, t, v) = \mathbb{E}[\varepsilon \mid x_t = x, v] = -\sqrt{v}\sigma(t)\nabla_x \log \tilde{p}_t(x \mid v). \quad (4.3.1.7)$$

4.3.2 Scale equivariance and standardised inputs

Proposition 29 (Scale equivariance). *Fix v and define the standardised process $y_t := (x_t - s(t)m_v)/\sqrt{v}$. Then y_t satisfies the v -free SDE*

$$dy_t = f(t)y_t dt + g(t)dB_t, \quad (4.3.2.1)$$

started from the standardised lifted conditional

$$q_0(y \mid v) \propto \mathcal{N}(y; 0, I_d) \Lambda(m_v + \sqrt{v}y), \quad \Lambda = p_{\text{data}}/p_{\text{wm}}. \quad (4.3.2.2)$$

Writing q_t for its marginals, the conditional score is equivariant,

$$\nabla_x \log \tilde{p}_t(x \mid v) = \frac{1}{\sqrt{v}} \nabla_y \log q_t \left(\frac{x - s(t)m_v}{\sqrt{v}} \mid v \right), \quad (4.3.2.3)$$

and all dependence on v beyond the affine standardisation is carried by the initial condition (4.3.2.2) alone.

Proposition 29 motivates parameterising the network in standardised coordinates,

$$\varepsilon_\theta(x, t, v) = u_\theta \left(\frac{x - s(t)m_v}{\sqrt{v}}, t, \log v \right), \quad (4.3.2.4)$$

so that the network receives sub-Gaussian $O(1)$ inputs however extreme the latent scale, and v enters only as a conditioning code. Together with Theorem 28, the lift stabilises the regression twice over: standard-normal targets and standardised inputs.

Corollary 30 (Exact-NMVM closed form). *If p_{data} coincides with the working-model marginal p_{wm} , then $\Lambda \equiv 1$ and (4.3.2.2) reduces to the v -free law $q_0 = \mathcal{N}(0, I_d)$; hence $q_t = \mathcal{N}(0, (s(t)^2 + \sigma(t)^2)I_d)$ and the optimal denoiser is independent of v in standardised coordinates. Under a variance-preserving schedule it is the linear map $\varepsilon^*(x, t, v) = \sqrt{1 - \bar{\alpha}_t} y_t$.*

Corollary 30 delimits when the standardised parameterisation (4.3.2.4) is free: at the exact-NMVM point the conditioning input is uninformative and u_θ need not use it. Away from it $q_0(\cdot | v)$ genuinely varies with v and the burden shifts onto the conditioning pathway. The effect is sharpest for multimodal targets, where a mode fixed at c_k in data space sits at $(c_k - m_v)/\sqrt{v}$ in standardised coordinates, so a network with limited conditioning capacity regresses towards an average geometry.

4.3.3 From training loss to generation error

Conditionally on v the setting is the classical one, so the standard bound for score-based samplers applies with no modification. Under Assumption 40, Girsanov’s theorem applied to the reverse SDE (4.2.2.1), in which v appears only as a constant in the diffusion coefficient, gives for each v

$$\text{KL}(\tilde{p}_0(\cdot | v) \| \hat{p}_0(\cdot | v)) \leq \text{KL}(\tilde{p}_T(\cdot | v) \| \pi_v) + \frac{1}{2} \int_0^T v g(t)^2 \mathbb{E}_{\tilde{p}_t(\cdot | v)} \|s_\theta - s^*\|^2 dt, \quad (4.3.3.1)$$

which is Theorem 41 (Chen et al., 2023, Thm. 2).

Two facts make (4.3.3.1) uniform over the lift rather than a family of unrelated bounds. The Girsanov weight is exactly the scale-adapted weighting of Theorem 28: by (4.3.1.5), $v g(t)^2 \|s_\theta - s^*\|^2 = (g(t)^2 / \sigma(t)^2) \|\varepsilon_\theta - \varepsilon^*\|^2$, whose scale is v -free, so a single likelihood-weighted excess risk controls every conditional at once. And by Proposition 29 the prior term equals its standardised counterpart, which vanishes with the terminal signal-to-noise ratio at the v -free rate of Proposition 21.

Mixing over $\tilde{p}(v)$ then reaches the target itself. Convexity of total variation bounds $\text{TV}(p_{\text{data}}, \hat{p}_0)$ by the $\tilde{p}(v)$ -average of the conditional distances, Pinsker’s inequality converts each into (4.3.3.1), and Proposition 11 is what makes p_{data} the correct mixture to compare against; this is Corollary 42. The two terms have separate origins; the integral is driven down by training (4.3.1.6) and the prior term by the schedule, so minimising the loss we train is minimising a bound on the error of the samples we generate.

4.4 Instantiation and practice

The latent is known for every training point but not for a sample we have yet to generate. Where does it come from then?

Lifting. By Proposition 12, the lift kernel is conjugate, so latents are sampled exactly. We resample $v^{(i)} \sim q(\cdot | x^{(i)})$ at each epoch, which draws fresh samples from the lifted joint $\tilde{p}$ rather than fixing a single augmentation—a finer Monte Carlo approximation of $\tilde{p}$ than a static lift, at no additional cost.

Training. The loss (4.3.1.6) is the standard noise-prediction loss with the two additions above; no other change to a standard diffusion codebase is required. Algorithm 1 summarises the procedure.

Generation. One draws v from the lifted latent marginal, initialises $x_T \sim \pi_v$ and integrates (4.2.2.1) or (4.2.2.2) with v frozen. By Theorem 16 the tail index of the generated law is a functional of the estimator $\hat{p}$ used at this step, so the choice of estimator is not an implementation detail. Table 4.1 sets out the three we compare in Section 6.5.1.

Estimator		Mechanism	Recovers	Costs
Pool	resam- pling	draws uniformly from the latents $\{v^{(i)}\}$ of the last lifting pass	the empirical latent law exactly	cannot exceed $\max_i v^{(i)}$, so the generated tail is truncated at that quantile
Fitted prior		fits a parametric density, e.g. a log-space mixture by EM whose E-step is the same conjugate posterior	a smooth density that interpolates between observed latents	imposes the tail of the chosen family, typically lighter than the data warrants
Peaks	over threshold	fits a generalised Pareto law to the exceedances $v^{(i)} - u_0$ above a high threshold, sampling the bulk empirically	regular variation beyond the observed range, consistent with Theorem 16	a threshold u_0 and a shape ξ to estimate, hence variance in the far tail

Table 4.1 Three estimators of the lifted latent marginal. Only the third is consistent with the regular-variation hypothesis that Theorem 16 needs, by the Pickands–Balkema–de Haan theorem (Balkema and de Haan, 1974; Pickands, 1975); the first two trade that away for, respectively, no estimation and a smooth density.

Instantiations. The framework is specified by the working model alone. Taking $\beta = 0$ with inverse-gamma mixing gives the Student- t lift used throughout Chapter 6; Proposition 12 covers the rest of the generalised hyperbolic family, and a positive $\alpha/2$ -stable mixing law gives the symmetric α -stable lifts by subordination, so the noise models of Chapter 5 appear as marginal views of a conditionally Gaussian process. For anisotropic data the lift is applied per coordinate with a product working model, replacing vI_d by $\text{diag}(v)$ and the isotropic conjugacy by its coordinatewise form.

Algorithm 1 Training with posterior lifting

Require: dataset $\{x^{(i)}\}_{i=1}^N$, working model (μ, β, p) , schedules $s(t), \sigma(t)$

- 1: **repeat**
- 2: **(re-)lift:** sample $v^{(i)} \sim q(\cdot | x^{(i)})$ for all i $\triangleright$ closed form, Proposition 12
- 3: **for** each minibatch $\{(x_0, v)\}$ **do**
- 4: sample $t \sim \mathcal{U}[0, T], \varepsilon \sim \mathcal{N}(0, I_d)$
- 5: $x_t \leftarrow s(t)x_0 + \sigma(t)\sqrt{v}\varepsilon$
- 6: $\hat{\varepsilon} \leftarrow u_\theta((x_t - s(t)m_v)/\sqrt{v}, t, \log v)$ $\triangleright$ standardised input, (4.3.2.4)
- 7: gradient step on $\|\hat{\varepsilon} - \varepsilon\|^2$ $\triangleright$ (4.3.1.6)
- 8: **end for**
- 9: **until** converged

Algorithm 2 Ancestral generation

Require: trained u_θ , latent pool $\{v^{(i)}\}$ or fitted prior $\tilde{p}(v)$, time grid $T = t_K > \dots > t_0 = 0$

- 1: sample $v \sim \tilde{p}(v)$ $\triangleright$ pool resampling, or fitted prior
- 2: sample $x_T \sim \pi_v = \mathcal{N}(s(T)m_v, v\sigma(T)^2I_d)$ $\triangleright$ (4.2.1.3)
- 3: **for** $k = K, \dots, 1$ **do**
- 4: $\hat{\varepsilon} \leftarrow u_\theta((x_{t_k} - s(t_k)m_v)/\sqrt{v}, t_k, \log v)$
- 5: $s_\theta \leftarrow -\hat{\varepsilon}/(\sqrt{v}\sigma(t_k))$ $\triangleright$ (4.3.1.5)
- 6: $x_{t_{k-1}} \leftarrow$ one solver step of (4.2.2.1) or (4.2.2.2) with score s_θ and v frozen
- 7: **end for**
- 8: **return** x_0 $\triangleright$ discard v ; marginal is p_{data} by Proposition 11

Chapter 5

Relationship to heavy-tailed noise models

The framework of Chapter 4 shares its starting point with the existing heavy-tailed generative models, namely the Gaussian scale-mixture representation of Section 2.3, and a natural objection is therefore that it is a reparameterisation of those models rather than a distinct one. This chapter answers that. We isolate the three choices that a scale-mixture construction must make, and for each one we identify which methods fall on which side. The design decisions are what the mixing variable attaches to (Section 5.2), whether the network is given it or it is marginalised away (Section 5.3), and whether it is drawn independently of the datum or inferred from it (Section 5.4). Only the third decision separates the lift from the rest.

5.1 A common template

Every construction considered here corrupts or interpolates the data with a kernel that is Gaussian once a mixing variable is fixed. Writing u for that variable, each method is an instance of the joint

$$p(x_0, u, x_t) = p_{\text{data}}(x_0) q(u | x_0) \mathcal{N}(x_t; m_t(x_0), c_t(u) I_d), \quad (5.1.0.1)$$

together with a rule stating whether the network receives u or only x_t . The instances are as follows.

- **t-EDM and t-Flow** (Pandey et al., 2024): $u = w \sim \text{InvGamma}(\nu/2, \nu/2)$ drawn afresh for each corruption, $q(u | x_0) = p(w)$, $m_t(x_0) = x_0$ and $c_t(w) = \sigma_{\text{EDM}}^2 w$. Marginalising w returns the Student- t kernel actually used.

- **DLPM** (Shariatian et al., 2025): $u = A_{1:T}$, a sequence of positive $\alpha/2$ -stable variables drawn independently at every step, $q(u | x_0) = \prod_t p(A_t)$, with c acting stepwise through $\mathcal{N}(x_t; \gamma_t x_{t-1}, \sigma_t^2 A_t I_d)$.
- **HTFM** (Yang et al., 2026): $u = T_{[0,1]}$, a nondecreasing path drawn from a prescribed clock law, $q(u | x_0) = \mathbb{P}_T$, $m_t(x_0) = a_t x_0$ and $c_t(T) = b_t^2 \Sigma(T_{[0,1]})$.
- **Posterior lift** (Chapter 4): $u = v$, drawn once per datum from the working-model posterior of Definition 10, $m_t(x_0) = s(t)x_0$ and $c_t(v) = v\sigma(t)^2$, giving

$$\tilde{p}(x_0, v, x_t) = p_{\text{data}}(x_0) \underbrace{q(v | x_0)}_{\text{posterior lift}} \mathcal{N}(x_t; s(t)x_0, v\sigma(t)^2 I_d), \quad v \not\perp x_0. \quad (5.1.0.2)$$

Under the Student- t clock $T_t = tV_v$ with $V_v \sim \text{InvGamma}(v/2, v/2)$ and the straight-line schedule, the area-clock covariance reduces to $\Sigma(T_{[0,1]}) = V_v I_d$, so the HTFM kernel is $\mathcal{N}(a_t x_0, b_t^2 V_v I_d)$. This is (5.1.0.2) with $(s, \sigma) = (a, b)$ after reversing the direction of time, and the conditioning input is $\log V_v$ in both cases (Yang et al., 2026, App. E.4). The two constructions therefore agree on the kernel, on the schedule family and on the conditioning code. What remains is the factor $q(u | x_0)$, and isolating the contribution that factor is the claim the rest of this chapter defends.

5.2 What the mixing variable attaches to

The first choice is which object carries the mixing variable. For t-EDM, t-Flow and DLPM it attaches to the *corruption*, scaling the noise added to the data. For HTFM it attaches to the *source*: the clock sets the covariance of the law the flow starts from, and the data endpoint is untouched. For the lift it attaches to the *datum*, with v drawn from the posterior given x_0 and the corrupted state inheriting it.

What this decides is whose tails conditioning tames. Attaching to the corruption or to the source leaves p_{data} as it was, so the tails the model must reproduce are still present in whatever the network regresses against. Attaching to the datum does not: by Proposition 14, $\tilde{p}(\cdot | v)$ is sub-Gaussian for every v , so the conditional learning problem is light-tailed on both ends.

HTFM makes the distinction visible in its own assumptions. Conditioning on the clock genuinely removes the moment condition on the *source*, which is what admits $v \leq 2$ and keeps the objective a plain L^2 regression. It removes nothing on the data side. Under the straight-line schedule the endpoint-conditioned velocity is $x_0 - X_0^T$ with X_0^T a centred Gaussian

conditionally independent of x_0 , so its conditional second moment is $\mathbb{E}\|x_0\|^2 + \text{tr}\Sigma(T_{[0,1]})$; the first term is not finite when p_{data} has infinite variance, in which case the square-integrability hypothesis of Yang et al. (2026, App. C.3) fails and Theorem 6 applies. The lift conditions on the other side, so Theorem 28 carries no moment condition on the data.

5.3 Marginalised or conditioned

The second choice is whether the network is given the mixing variable. Here the four constructions split two against two, and the split predicts a great deal.

Marginalising. t-EDM, t-Flow and DLPM integrate the mixing variable out, and LIM (Yoon et al., 2023) dispenses with the augmentation altogether. The law the network faces is then heavy-tailed and the squared loss is unavailable: t-EDM and t-Flow replace it with the γ -power divergence (Eguchi, 2021), DLPM with a fractional L^r norm, LIM with a fractional score.

Marginalising also constrains the backbone, because the heavy-tailed law must support the way the forward process is specified. By Section 2.3.2, composing a transition kernel yields a closed-form perturbation kernel only for a convolution-closed family. The Gaussian and α -stable families are closed, the latter being the stability property Shariatian et al. (2025) invoke to telescope the DLPM chain; the Student- t family is not. A t-DDPM would therefore have no closed-form perturbation kernel, and hence no closed-form denoising posterior. The closed form for the γ -power divergence (Pandey et al., 2024, Eq. 7) also holds only between Student- t laws sharing a common v , so the per-step term would not reduce to a weighted squared error. This is why Pandey et al. (2024) instantiate on EDM and on the straight-line flow, both of which specify the perturbation kernel directly and never compose increments. The obstruction is thus just how the forward process is specified.

Conditioning. HTFM and the lift give the mixing variable to the network. Every object it then faces is the classical Gaussian one, so the plain squared loss applies unmodified and the noise family constrains no backbone: conditionally on the mixing variable the increments are Gaussian, so both specification styles are available. For the lift, (4.2.1.1) is linear and Brownian-driven, so a DDPM-style ancestral chain is available; equally the conditional perturbation kernel of Lemma 20 may be specified directly through (s, σ) , giving EDM and straight-line-flow instantiations, while (4.2.2.1) and (4.2.2.2) share marginals as in the Gaussian theory and a DDIM-style sampler needs no modification.

This freedom belongs to conditioning: Yang et al. (2026, App. A.5, App. B) give variance-preserving and variance-exploding clock-conditioned paths and extend the clock to diffusion models with reverse kernels and a conditional denoising objective.

5.4 Independent or coupled

The third choice is whether $q(u | x_0)$ depends on x_0 . Here the split is one against the rest.

Coupling is a statement about independence, so no reparameterisation can introduce or remove it: no change of variables applied to a joint in which $u \perp\!\!\!\perp x_0$ produces one in which u carries information about x_0 . HTFM’s velocity field is conditioned on the mixing object, and for the Student- t clock that object is the scalar $\log V_v$, exactly as in (4.3.2.4); yet the clock is drawn from a prescribed law. Conditioned and coupled are separate properties, and only the first has been exploited.

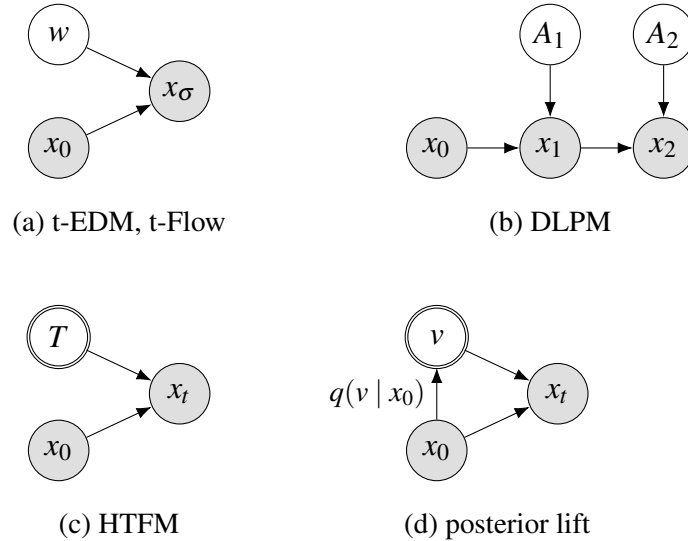


Figure 5.1 The same scale-mixture primitive placed at four points of the generative graph. Shaded nodes are data or corrupted states, plain circles are mixing variables that are marginalised and never observed, double circles are mixing variables supplied to the network. The variable is drawn once per corruption in (a) and once per step in (b), independently of the data in both. In (c) the clock is given to the velocity field but still drawn from a prescribed law. Panel (d) adds the edge $x_0 \rightarrow v$, which is the whole of the structural difference.

The edge is observable, and the following statistic shows how.

Definition 31 (Conditional noise scale). For a forward process whose corrupted state has conditional covariance $\text{Cov}(x_t | x_0) = c(t) \zeta(x_0) I_d$ for some schedule-dependent $c(t) > 0$, we call $\zeta(x_0)$ the *conditional noise scale* of the process at x_0 .

Lemma 32 (Discriminating lemma). *Let p_{data} be arbitrary and consider the template (5.1.0.1).*

1. *If $u \perp\!\!\!\perp x_0$, then $q(u \mid x_0) = p(u)$ and the conditional noise scale is constant in x_0 wherever it exists. This covers t -EDM, t -Flow, DLPM and HTFM. For the Student- t instances it equals $\mathbb{E}[w] = \nu/(\nu - 2)$, and hence requires $\nu > 2$.*
2. *Under the posterior lift (5.1.0.2) with $\text{GIG}(\lambda, \chi, \psi)$ mixing, the conditional law of ν is the posterior (4.1.2.1) and depends on x_0 through $\|x_0 - \mu\|^2$. For the Student- t lift,*

$$\zeta_{\text{lift}}(x_0) = \mathbb{E}[\nu \mid x_0] = \frac{\nu + \|x_0 - \mu\|^2}{\nu + d - 2}, \quad \nu + d > 2, \quad (5.4.0.1)$$

which is strictly increasing and unbounded in $\|x_0 - \mu\|$.

3. *Hence the two families of forward process have different conditional covariance structures, constant against affine in $\|x_0 - \mu\|^2$, and are distinct stochastic processes.*

The proof is given in Appendix A. Lemma 32 assumes no mismatch between the data and the working model: it holds for arbitrary p_{data} , in particular when p_{data} is exactly the Student- t these frameworks are built to model, and because part (i) is a statement of independence no reparameterisation of an uncoupled joint can reproduce (5.4.0.1). It also supplies the diagnostic reported in Section 6.4: regressing the empirical conditional variance of x_t on $\|x_0 - \mu\|^2$ gives a flat profile for the uncoupled constructions and a linear one of slope $(\nu + d - 2)^{-1}$ for the lift.

What coupling buys is that the mixing law becomes a functional of the data. Where $u \perp\!\!\!\perp x_0$ the law of u is chosen in advance, and so is the tail weight of what the model generates. Tail recovery is then a tuning problem with an unknown target. Under the lift the mixing law is $\tilde{p}(\nu)$, which by Corollary 18 carries the tail index of the data whatever working ν is used, and asymmetry follows from β in the same way; the mean-variance extension sketched by Yang et al. (2026, App. A.4) makes its mean depend on the clock rather than on the datum, so it cannot represent data-dependent skewness.

This placement has been identified as the natural next step by Yang et al. (2026, App. A.2). They propose inferring a data-dependent posterior over clock paths with an amortised model, which would turn the mixing object into a learned description of local tail behaviour rather than a prescribed family, and defer it because such inference needs further identifiability and estimation assumptions. That reservation is appropriate for a learned, path-valued latent. It does not apply to a scalar mixing variable drawn from a conjugate working model, where the posterior is available in closed form (Proposition 12) and nothing is estimated.

5.5 Consequences for modelling capability

	t-EDM / t-Flow	DLPM	HTFM	Posterior lift (ours)
Attaches to	corruption	increment	source	datum
Drawn	per corruption	per step	per sample	per datum, frozen
Coupled to x_0	no	no	no	yes, via $q(v x_0)$
Seen by network	no	no	yes	yes
Forward specified by	perturbation kernel	transition kernel	perturbation kernel	either
Backbone	EDM, straight-line flow	DDPM; DDIM-style DLIM	flow; VP/VE and diffusion extensions	VP/VE, discrete or continuous, SDE or ODE
Regression target	heavy-tailed	heavy-tailed	Gaussian given the clock	standard normal
Loss modification	γ -power divergence	$L^r, r < 2$	none	none (plain MSE)
Condition on the noise or source	$v > 2$ for t-EDM	$r < \alpha/2, \alpha \in (1, 2)$	none, any $v > 0$	none
Condition on the data	avoided by the divergence	avoided by the L^r norm	finite variance for the L^2 objective	none
Skewness	none	none	clock-dependent mean, not pursued	native, via β
Generated tail index	v , tuned	α , tuned	clock law, tuned	inherited from data

Table 5.1 The four constructions read through the template (5.1.0.1). The first three rows are the choices of Sections 5.2 to 5.4 and the remainder are their consequences. The two moment rows are kept apart because they concern different objects: the first is a condition on the corruption or source law and the second is a condition on p_{data} . Conditioning removes the loss modification and frees the backbone; only attaching to the datum removes the condition on the data, and only coupling makes the tail index a functional of it.

Table 5.1 collects the consequences. The absence of a condition on the data is not one property shared by the last two columns, because different mechanisms produce it.

A condition may be *avoided*, by replacing the objective with one insensitive to the heavy tail. This is what the γ -power divergence does for t-EDM and t-Flow and the fractional L^r norm for DLPM. Shariatian et al. (2025) are explicit that their network models a light-tailed conditional, but since the network is not given the mixing variables the target is light-tailed only conditionally on them.

Or a condition may be *removed*, in the sense that the regressed quantity has no heavy tail to condition on. Under the lift the target is $\varepsilon \sim \mathcal{N}(0, I_d)$ exactly (Theorem 28), so the squared loss is finite because of what is regressed rather than how residuals are weighted. The distinction has a cost attached: a robust divergence downweights the largest residuals, and under a heavy-tailed target those are contributed by the events the model exists to represent, so robustness is in tension with the estimand. Under the lift no such trade is made, the extremeness having been routed into $\tilde{p}(v)$, which is never score-matched.

HTFM removes the condition in the source and retains the one in the target.

Chapter 6

Evaluation methodology and experiments

Does routing the tails into a latent that is never learned actually recover them, and where does the construction stop paying?

Section 6.1 constructs the targets. Section 6.2 develops an evaluation protocol, since the statistics used to compare generative models are not well defined for the targets under our study. Then, Section 6.3 describes the models and the training budget, and Section 6.4 verifies that each implemented forward process is the one its paper describes. The remaining sections report results. Numerical details omitted here are collected in Appendix B.

6.1 Targets and data

The synthetic targets are Student- t normal mean–variance mixtures,

$$x = c_k + \beta v + \sqrt{v} z, \quad v \sim \text{InvGamma}(v/2, v/2), \quad z \sim \mathcal{N}(0, I_d), \quad (6.1.0.1)$$

with one scalar v per sample shared across dimensions, so the per-dimension law is Student- t and the tail index is exactly v . Table B.1 lists them. The field targets share a per-pixel marginal and differ only in how spatially coherent their extremes are, which makes them a misspecification ladder whose rungs are fixed by construction and invisible to any marginal statistic.

Real data comes from the High-Resolution Rapid Refresh (HRRR), NOAA’s hourly-updating convection-allowing analysis of the conterminous United States on a 3 km grid (Dowell et al., 2022). We use the two channels of Pandey et al. (2024): vertical velocity at hybrid level 20 (w20), and vertically integrated liquid (VIL). Analysis crops over the Central Plains for 2019 to 2021 are split chronologically, with 2021 held out, and each 128×128

Family	Construction	χ_4	Exceedance area	Working model
Scalar ν	$x(s) = \sqrt{\nu}z(s)$, one ν per field	0.28	8.8%	exact
Local scale, $\ell_\nu = 4$	$\nu(s)$ a smooth field with exact inverse-gamma marginal	0.22	6.0%	approximate
Local scale, $\ell_\nu = 2$	as above, rougher scale field	0.08	2.1%	approximate
Gaussian copula	Student- t margins, asymptotically independent extremes	0.07	2.1%	structurally wrong
Skewed copula	as above with skew- t margins, $\gamma = 2$	0.07	2.1%	structurally wrong

Table 6.1 The field targets, ordered by the spatial coherence of their extremes. The per-pixel tail index is the same throughout, so the families differ only on the axis measured by χ_4 and the exceedance area (Section 6.2), which spans a factor of four. The scalar- ν family is an exact normal mean–variance mixture; the copula families admit no scalar or field ν with $x = \sqrt{\nu}z$, so there the working model is structurally wrong. All families but the skewed copula also have a $\nu = 1.5$ variant.

hour is tiled into sixteen disjoint 32×32 crops rather than block-averaged, since averaging is a central-limit operation that lightens the tail and so deletes the phenomenon under study.

Normalisation differs by target and is recorded in Table B.2.

6.2 Evaluating generative models of heavy-tailed data

Pandey et al. (2024) compare generated and reference samples through kurtosis and skewness ratios. At the tail indices studied here those statistics do not converge.

Proposition 33 (Non-convergence of sample moment ratios). *Let $X_1, \dots, X_n$ be i.i.d. with $|X| \in \text{RV}_{-\alpha}$ and let $g_1^{(n)}, g_2^{(n)}$ be the sample skewness and kurtosis. Then $g_1^{(n)} \rightarrow \infty$ in probability if $\alpha < 3$ and $g_2^{(n)} \rightarrow \infty$ if $\alpha < 4$, so the ratios formed from two such samples have no deterministic limit.*

Our synthetic targets reach $\alpha = 1.5$ by construction. Every scored quantity is therefore built from quantiles, ranks or the tail index. The ratios are still computed on the real data for comparability with their tables, but enter no aggregate.

The four quantities that carry the results are the Hill estimator on the top k order statistics $r_{(1)} \geq \dots \geq r_{(k)}$ of the radius, its z -score, the integrated log-quantile error, and the composite:

$$\hat{\alpha}_k = \left(\frac{1}{k} \sum_{i=1}^k \log(r_{(i)}/r_{(k)}) \right)^{-1}, \quad (6.2.0.1)$$

$$z = \frac{|\hat{\alpha}_{\text{gen}} - \hat{\alpha}_{\text{ref}}|}{\sqrt{\hat{\alpha}_{\text{ref}}^2/k_{\text{ref}} + \hat{\alpha}_{\text{gen}}^2/k_{\text{gen}}}}, \quad (6.2.0.2)$$

$$\text{MSLE} = \int_{0.95}^1 (\log F^{-1}(p) - \log \hat{F}^{-1}(p))^2 dp, \quad (6.2.0.3)$$

$$C_{\mathcal{S}} = \exp \left(\frac{1}{|\mathcal{S}|} \sum_{m \in \mathcal{S}} \log \max \{M_m/f_m, 1\} \right), \quad (6.2.0.4)$$

with $k = 0.02n$ throughout, F^{-1} and $\hat{F}^{-1}$ the reference and generated quantile functions, f_m the noise floor of metric m and M_m the model's value. So $z \approx 1$ means two tail indices indistinguishable at this sample size, and $C_{\mathcal{S}} = 1$ means indistinguishable from an independent draw of the target on every metric in $\mathcal{S}$. Table 6.2 gives the full suite and Table 6.3 the conventions governing every comparison.

The floor f_m is the median of metric m over five independent reference pairs at the sample sizes used in the comparison. On HRRR there is no generator, so the pairs are drawn from the training years and from the test year and the floor measures interannual shift instead of sampling noise: the two years differ enough that several models score below it, so Section 6.7 leads with raw metrics.

Two entries of Table 6.2 need their definitions: the extremal dependence coefficient is $\chi(h) = \mathbb{P}(X_{i+h} > u_0 \mid X_i > u_0)$ at a high threshold u_0 , reported at lag $h = 4$, and the exceedance area is the expected fraction of pixels above u_0 given at least one exceedance. Three of these would change a conclusion if set differently. Radii about a single centre on a multimodal target place the bulk at the inter-mode distance, so Hill reads the bulk shoulder rather than the tail. Selecting the numbers reported on the panel amounts to fitting the sample noise, increasing the tail composite error by a factor of two for one vector target. And on the skewed field both tails share the same index by construction, so an index-based rule for deciding which side to score is a coin flip; forcing the lighter side increases the pooled MSLE by a factor of 55.

6.3 Models, architectures and budget

We compare DDPM (Ho et al., 2020), t-EDM and t-Flow (Pandey et al., 2024), DLPM (Shariatian et al., 2025) and the posterior lift of Chapter 4. On the HRRR branches we drop

Metric	Measures	Reported on
Hill index (6.2.0.1)	tail index of the radius	per-sample and pooled pixels
Hill z (6.2.0.2)	whether two indices differ	per-sample only
Tail KS	distance beyond the reference 0.99 quantile	both; sentinel 1 on total collapse
MSLE (6.2.0.3)	log-quantile error over the upper tail	both
W_1	bulk fidelity	both; sliced over 128 projections above $d = 1$
Mode weights, inter-mode mass	bulk fidelity	multimodal targets only
$\chi(h)$, exceedance area	spatial coherence of extremes	diagnostic, never aggregated
Negative fraction	mass below zero before clamping	diagnostic, VIL only

Table 6.2 The metric suite, computed in raw denormalised coordinates. Radii are measured to the nearest known mode centre. Composites are formed separately over tail and bulk sets and over the per-sample and pooled families. Undefined metrics are skipped.

Convention	Rule
Per-model tuning	each model swept over its own tail hyperparameter on every target, all grids over the same seeds and the same selection rule
Which tail is scored	read from the reference, never from the generated sample
Budget	architecture and budget identical within an experiment, sampling matched by network evaluations (NFE)
Seeds	five on synthetic targets, three on HRRR, as median and interquartile range

Table 6.3 Conventions governing every comparison.

DLPM due to computing budget constraints, and add EDM (Karras et al., 2022) and Gaussian flow matching as light-tailed controls for the two Student- t constructions. Tables B.3 and B.4 give the architectures, budgets and samplers, and the accompanying code release gives the full configurations. HTFM (Yang et al., 2026) is not among the baselines simply because its preprint appeared too close to the dissertation submission deadline to implement and test it. Chapter 5 treats it analytically, as the construction closest to the lift.

Because DLPM performs poorly on several targets, we ran the authors’ own code on shared inputs (schedule, forward process, prediction target, loss and posterior moments); the two agree exactly.

6.4 Validating the forward processes

Each kernel is tested against a closed form it must satisfy exactly, with no network involved, by binning held-out data on $r^2 = \|x_0 - \mu\|^2$ and measuring the conditional variance of the corrupted state with the schedule divided out. Binning is what makes the test discriminating: by Lemma 32 the uncoupled kernels must give a flat profile and the lift an affine one. Appendix B gives the statistic, the estimators and the full grids.

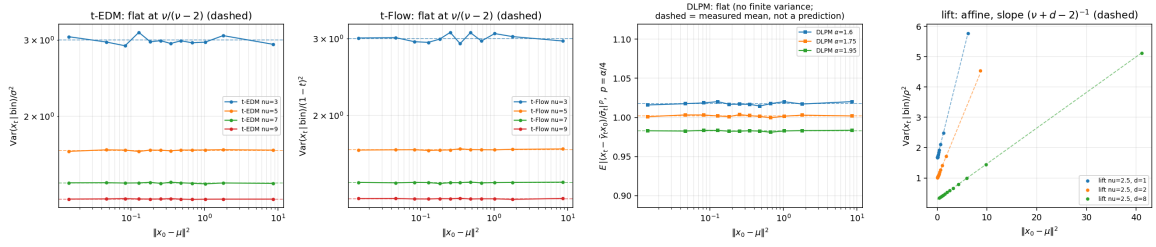


Figure 6.1 Each forward process against the closed form it must satisfy, on a held-out Student- t target with $v = 2.5$. The rightmost panel is the coupled–uncoupled distinction of Chapter 5 measured in the corruption process itself, before any sample is generated

All four agree with their closed forms, so differences below can be attributed to the methods.

6.5 Tail-index recovery

Corollary 18 claims that the tail index of the generated law is inherited from the data and does not depend on the working degrees of freedom. In this section, we check this.

Target	d	α_{ref}	working v	$\alpha_{\text{gen}}, \beta_{\text{work}} = 0$	$\alpha_{\text{gen}}, \beta_{\text{work}} = \beta_{\text{data}}$
Field, $\ell_v = 2$	256	1.452	1.0 to 1.5	1.426 to 1.450	—
Skewed vector	8	1.489	1.5 to 2.5	1.283 to 1.303	1.402
Skewed vector	32	1.460	1.5 to 2.5	0.728 to 0.846	0.980
Misspecification	8	2.175	1.5 to 20	2.008 to 2.372	2.179
Misspecification	32	2.260	1.5 to 20	1.680 to 1.761	2.308

Table 6.4 Hill index of generated samples as the working degrees of freedom are swept at fixed data, medians over five seeds. The width of the fifth column is what Corollary 18 predicts should be small; the sixth repeats it with the working skewness set to the data's. Only the working model changes

Sweeping the working v by a factor of thirteen moves the generated index by 18% at $d = 8$ and 5% at $d = 32$ on the misspecification targets, and by under 2% on the skewed

vector target at $d = 8$, though over a sweep of only 1.67 there (1.5 to 2.5), and on the field target. The index is a property of the data rather than of the working model, which is the claim.

Recovering it at the right *level* depends on the working skewness instead. A symmetric working model recovers the field target to within 2% but undershoots the skewed vector target at $d = 32$ by more than a third, and setting β_{work} to the data's value recovers about a third of the shortfall there and all of it on the misspecification target. The shortfall is not in the lift, whose latent pool carries the index predicted by Proposition 12 to within noise, but in the learned conditional at large v where training data is sparse, which is the conditioning-capacity cost Corollary 30 anticipates.

6.5.1 Estimating the lifted latent marginal

By Theorem 16, the generated index is a functional of the law v is drawn from at generation, so the estimators of Table 4.1 have a predicted consequence: on the $l_v = 2$ field at $v = 1.5$ the pool has index 0.705 and generates 1.426 against the 1.410 its pool index implies, while the fitted tail has 0.955 and generates 1.905 against an implied 1.910.

Which is better is target-dependent. Peaks-over-threshold wins on every vector target, taking the tail composite from 13.19 to 6.45 at $d = 32$; pool resampling wins where the chain above is faithful, sitting at the sampling floor on the scalar- v field where extrapolation moves it to 3.85. Read together, extrapolation is not repairing anything: it lightens the latent under almost any configuration and helps only where the learned conditional makes the tail too heavy, so two errors partly cancel.

6.6 Regime map: where the lift wins and where it pays

Inheriting the tail index is one property among many. What does it cost elsewhere?

6.6.1 Vector targets

Table 6.5 reports the vector targets. The grid mixture is the informative case. At $d = 2$ every working v gives $v + d \leq 3.5 < 4$, so by Remark 37 the conditional scale has no variance; the generated index is then too light, 2.37 against a reference of 1.53, and the independence from v degrades, α_{gen} moving by a quarter as v goes from 1.0 to 1.5 where at $d = 32$ a factor of thirteen moved it by a twentieth.

Model	Skewed, $d = 8$		Skewed, $d = 32$		Grid mixture, $d = 2$	
	tail	bulk	tail	bulk	tail	bulk
DDPM	5.02	6.19	5.39	11.16	10.05	16.13
DLPM	6.08	3.79	14.15	6.73	2.97	5.98
t-EDM	3.11	1.79	6.15	4.31	3.50	17.32
t-Flow	3.37	1.10	21.40	37.62	3.08	14.25
Posterior lift	6.69	2.09	6.45	1.87	5.20	15.36

Table 6.5 Radius-family composites on the three vector targets, medians over seeds at each model’s best configuration. The lift is behind on every tail column (Section 6.5 locates the cause) and ahead on the bulk at $d = 32$, where it nonetheless recovers the tail index most accurately of any model.

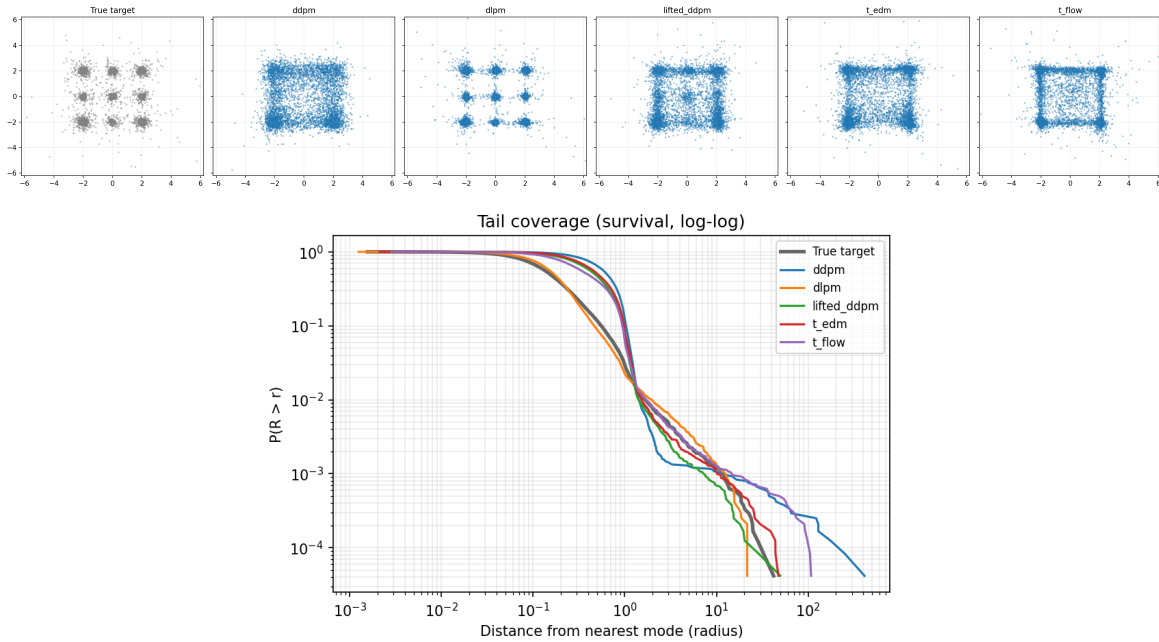


Figure 6.2 The grid mixture at $v = 1.5$: samples above, survival function of the distance to the nearest mode below. Only DLPM reproduces the nine modes, and the inter-mode leakage is 0.003 for DLPM against 0.035 for the lift.

Figure 6.2 shows that mode structure is lost by every model except DLPM. That failure is not intrinsic to the lift.

6.6.2 Working-model misspecification

Table 6.7 gives the content of exactness in practice. The comparison that supplies it is with the baselines when their own tail parameter is wrong: on a comparable skewed target, t-Flow reaches a tail composite of 21.4 and a bulk composite of 37.6, and DLPM 14.2. Their tail

Centring for the lift	tail	bulk	inter-mode leakage
Nearest of nine k -means centres	7.56	7.39	0.004
Single global centre	5.20	15.36	0.035

Table 6.6 Centring ablation on the grid mixture. Supplying the mode locations improves the bulk and the leakage, but shrinks every radius and so lightens the generated tail; since the lift under-produces the tail here, global centring partly compensates for it. We report both.

parameter is not a working assumption that costs accuracy but a specification of the model that, set wrongly, breaks it. Exactness (Proposition 11) is what turns a specification error into an estimation cost for our model.

Working model	Error introduced	tail	bulk	α_{gen}
$\nu = 5, \beta = 1$	none	1.03	2.16	2.31
$\nu = 5, \beta = 0.5$	skewness halved	1.30	2.25	2.23
$\nu = 5, \beta = 0$	skewness removed	2.56	2.63	1.74
$\nu = 1.5, \beta = 0$	and ν too light	2.41	2.30	1.76
$\nu = 20, \beta = 0$	and ν too heavy	2.86	3.25	1.68
GIG mixing, $\beta = 0$	wrong mixing family	1.94	2.11	1.90

Table 6.7 Deliberate misspecification on the $d = 32$ target, reference index 2.26. Proposition 11 concerns the estimand and cannot be measured directly, so what is reported is whether the estimate degrades gracefully: across a working ν wrong by a factor of thirteen and a skewness set to zero on data with $\beta = 1$, the tail composite moves by under a factor of three. The last row lies outside Assumption 13 and is excluded from any claim.

6.6.3 Field targets

The lift wins every cell, including the rungs where its working model admits no scalar or field ν . This is the clearest practical evidence for Proposition 11: the working model may be wrong about how the field is generated and still target the right law.

The cost is in $\Delta\chi_4$. The lift overstates the spatial coherence of extremes, on average by 20% on the two copula families against under 4% for t-EDM, and most where the truth is asymptotically independent: a single ν scales the whole field, so a large draw raises every pixel together and extremes are coherent by construction. Where a scale field genuinely exists, the lift has the smallest coherence error of any model, and attaching one latent per block, which the coordinatewise form of Proposition 12 supports, is the natural remedy. The ladder is also where Remark 43 bites: at $\nu = 1.5$ the lift is swept over working $\nu \in \{1.0, 1.25, 1.5\}$,

Rung	$\nu = 2.5$			$\nu = 1.5$		
	Lift	Best baseline	$\Delta\chi_4$	Lift	Best baseline	$\Delta\chi_4$
Local scale, $\ell_\nu = 2$	3.31	5.43	−1%	5.64	18.91	+5%
Local scale, $\ell_\nu = 3$	3.78	8.52	+10%			
Local scale, $\ell_\nu = 4$	3.97	7.49	+1%			
Scalar ν	1.88	3.11	+8%	1.53	7.12	+1%
Gaussian copula	2.82	4.29	+16%	3.26	7.55	+23%
Skewed copula	1.94	4.94	+23%			

Table 6.8 The field ladder of Table 6.1 at two tail indices. Lift and baseline columns are pooled-pixel tail composites, medians over five seeds; $\Delta\chi_4$ is the relative error in extremal dependence at lag four and enters no composite. Blank cells were not run.

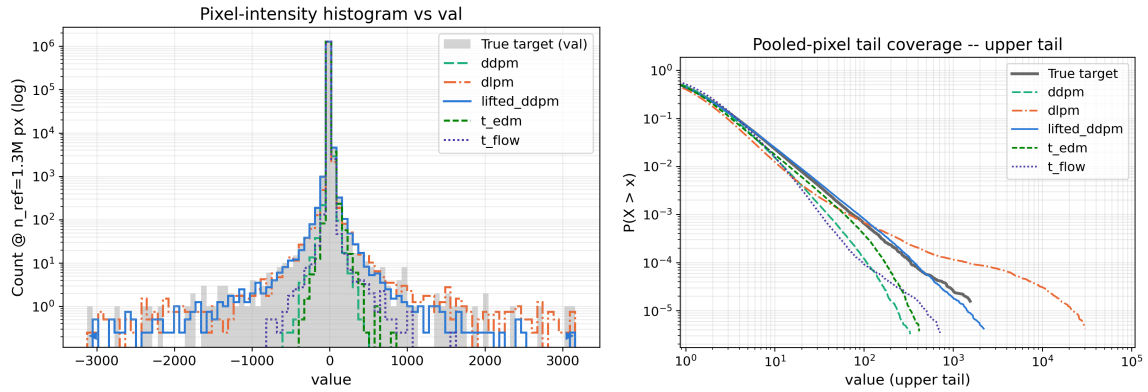


Figure 6.3 The Gaussian copula field at $\nu = 1.5$, where the working model is structurally wrong. Pooled pixel intensities left, per-sample radius right. The lift follows the reference across three decades and truncates only at the last order statistic, while the Student- t constructions fall away near the 10^{-3} exceedance level and DLPM overshoots by more than an order of magnitude.

none of which t-EDM’s preconditioning admits, so its grid starts at 2.5, and its generated index is 2.84 against a reference of 1.45.

6.7 Real data

6.7.1 Vertical velocity (w20)

Table 6.9 confirms the chapter’s claims: the lift leads the best baseline by a factor of 2.44 on the tail composite, with the same ordering on both reference panels and at all sampling budgets. Two readings separate. The split on the tail index is by noise law rather than by parameterisation, since EDM predicts x_0 and Gaussian flow predicts the noise, and both sit

Model	pix-tail composite	pix-MSLE (floor 7.2×10^{-5})	pix-Hill error (floor 0.167)	pix- W_1 (floor 4.6×10^{-4})
Posterior lift	1.08 [1.05, 1.22]	4.8×10^{-5}	0.183	0.00384
t-EDM	2.63 [2.26, 3.75]	4.8×10^{-4}	0.274	0.00300
t-Flow	3.87 [2.79, 5.66]	7.5×10^{-4}	0.308	0.00333
DDPM	5.12 [5.06, 5.21]	1.9×10^{-3}	0.827	0.00443
EDM	5.31 [3.66, 6.70]	3.3×10^{-3}	0.540	0.00297
Gaussian flow	5.92 [4.52, 8.89]	2.9×10^{-3}	1.251	0.00351

Table 6.9 HRRR w20, medians over three seeds with interquartile range, selected on the training panel and reported on the test year. The lift is separated from every baseline by interquartile range; no two adjacent baselines are separated at three seeds. The pooled Kolmogorov–Smirnov term is omitted because it saturates.

on the wrong side of it. And t-EDM recovers the slope while placing it at the wrong level, second on the Hill error and ten times worse on the quantile error.

The bulk is partly traded here, which the synthetic fields did not show: the lift is fifth of six on pooled W_1 . The bulk spans a factor of 1.5 across all six models where the tail spans 5.5, and the selection metric is tail-only, so it cannot have traded one for the other; a lift configuration at 1000 NFE reaches 5.4×10^{-4} , five times better than any other model at any budget, and selection does not choose it. The synthetic claim that the tail is won without cost to the bulk does not transfer to w20, and we do not state it generally.

6.7.2 Vertically integrated liquid (VIL)

The VIL floor is a train-against-test comparison between different climates, the training years having a 99.9th percentile of 38.9 against the test year’s 14.2. Every floor is one to three orders of magnitude larger than on w20, half of all runs clip on the tail statistics, and the composite table spans a factor of 1.52, inside the seed spread of five of six models, so we report raw columns instead.

The two panels then say opposite things, but only one is interpretable. Against the training distribution, the lift is first on all three tail statistics, which is the ordering w20 and the synthetic suite both produce. Against the test year, DDPM leads and no consistent ordering exists, because 2021 is a much lighter VIL year, so under-reproducing the training tail moves a model towards the test distribution.

Figure 6.4 shows that failure matters more than the ranking, and every quantile and rank statistic in Section 6.2 is invariant to it: a diffusion model on unbounded coordinates cannot represent this variable’s support, and that dominates any tail comparison on VIL. A

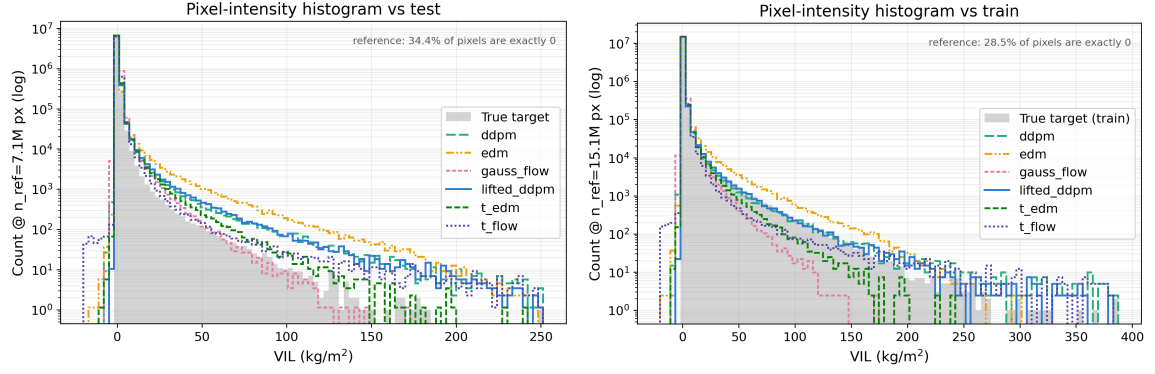


Figure 6.4 Pooled pixel intensities on HRRR VIL, against the test year (left) and the training years (right). The reference has an atom at zero holding 34.4% of its mass, at which no model places a single pixel, and every model emits negative values, from 12.7% for t-EDM to 56.8% for t-Flow, with the lift at 29.4%.

support-respecting parameterisation is the obvious response, and we have not tested one. The kurtosis and skewness ratios of Pandey et al. (2024) are reported in Appendix B for comparability, their train-against-test floor reaching 1.028, which is two divergent quantities behaving as Proposition 33 says they must.

Model	<i>pix</i> -tail composite	<i>pix</i> -MSLE (floor 1.38×10^{-2})	<i>pix</i> -Hill error (floor 0.331)	<i>pix</i> - W_1 (floor 0.189)
t-EDM	1.00 [1.00, 1.09]	0.00978	0.218	0.1376
Posterior lift	1.07 [1.03, 1.16]	0.01274	0.319	0.2340
DDPM	1.14 [1.07, 1.25]	0.00416	0.370	0.1365
Gaussian flow	1.18 [1.09, 1.21]	0.00710	0.163	0.2235
t-Flow	1.33 [1.16, 4.39]	0.03228	0.118	0.2844
EDM	1.52 [1.28, 1.63]	0.03202	0.126	0.3345

Table 6.10 HRRR VIL, medians over three seeds with interquartile range, selected on the training panel and reported on the test year. The composite does not separate the models here due to interannual shift. Five of six models overlap in IQR with the leader, and t-EDM’s 1.00 is just metric saturation.

6.8 Limitations

- Dimension and architecture: every target on which the lift recovers the index cleanly is high-dimensional and convolutional; every target on which it falls short is low-

dimensional and a perceptron, so the diagnosis of Section 6.5 is established for these targets rather than shown to be a property of dimension;

- Both HRRR channels come at a quarter of the resolution and a fifteenth of the training budget of Pandey et al. (2024), so only the ordering within this benchmark is comparable with theirs;
- DLPM is absent from the real-data experiments and from the budget sweep due to computing budget constraints, and the nearest competing construction, HTFM, is compared only analytically (for the reason given in Section 6.3).

Chapter 7

Conclusion & future work

The posterior lift routes the heavy tails of the data into a latent scale that the diffusion never touches. Conditioning on that scale leaves the target unchanged, by Proposition 11, and makes every conditional sub-Gaussian, by Proposition 14, so the classical diffusion framework applies with one substitution. The tail index of the generated law is then a functional of the data rather than a hyperparameter, which Theorem 16 and Corollary 18 establish and which empirical sweeps confirm.

The experiments locate both the reach and the price. The lift wins every rung of the misspecification ladder, including those where its working model admits no latent scale at all, and by a widening margin as the tail heavies; on real vertical-velocity data it leads the best baseline by a factor of 2.44 on the tail composite. It pays in three places, each traceable to the construction rather than to the implementation. A single latent scales the whole field, so extremal dependence is overstated where the truth is asymptotically independent. At $d = 2$ the conditional scale has no variance and the guarantees degrade. And where the generated tail falls short of its prediction, the cause is the learned conditional at large v , where training data is sparse, rather than the lift itself.

Four directions follow directly. A block-wise latent, which the coordinatewise form of Proposition 12 already supports, would address the coherence overstatement, with the granularity as the open question, since finer blocks weaken the identifiability of each latent. A support-respecting parameterisation is needed for bounded or non-negative variables, where no model tested here places mass on the zero atom or avoids negative values. Hypothesis (H) of Section A.2 should be discharged, since only the index is claimed and not the sharpness of the slowly varying factor. And the coupling developed here is scalar and inferred in closed form, whereas Yang et al. (2026) raise a data-dependent path-valued clock as future work and defer it on identifiability grounds; whether the conjugate route extends to that setting is the natural question to ask next.

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Appendix A

Proofs

A.1 Proofs for Chapter 3

Theorem 6. *(The L^2 objective is ill-posed) The field $b^\star$ is the minimiser of the regression risk*

$$\mathcal{L}(b) = \mathbb{E} \|b(t, x_t) - (x_1 - x_0)\|^2. \quad (3.2.0.1)$$

If either endpoint has infinite second moment, then $\mathcal{L}(b) = \infty$ for every b with $\mathbb{E} \|b(t, x_t)\|^2 < \infty$, and the objective has no finite minimiser. This holds independently of any noise added to the interpolant, since the regression target $x_1 - x_0$ is intrinsic to (3.1.0.1).

Proof. For any measurable b , condition on x_t and decompose the risk (3.2.0.1) by the conditional bias–variance identity,

$$\mathcal{L}(b) = \mathbb{E} \|b(t, x_t) - b^\star(t, x_t)\|^2 + \mathbb{E} \text{Var}[x_1 - x_0 \mid x_t],$$

where $b^\star(t, x) = \mathbb{E}[x_1 - x_0 \mid x_t = x]$ and the cross term vanishes because $x_1 - x_0 - b^\star$ is conditionally centred. The second term is intrinsic to the interpolant and independent of b . Now

$$\mathbb{E} \text{Var}[x_1 - x_0 \mid x_t] = \mathbb{E} \|x_1 - x_0\|^2 - \mathbb{E} \|b^\star(t, x_t)\|^2.$$

For any b with $\mathbb{E} \|b(t, x_t)\|^2 < \infty$, the elementary inequality $\|u - w\|^2 \geq \frac{1}{2} \|w\|^2 - \|u\|^2$ with $w = x_1 - x_0$ gives

$$\mathcal{L}(b) \geq \frac{1}{2} \mathbb{E} \|x_1 - x_0\|^2 - \mathbb{E} \|b\|^2 = \infty$$

whenever either endpoint has infinite second moment.

The argument uses only the target $x_1 - x_0$ of (3.2.0.1), which is fixed by the interpolant (3.1.0.1) and unaffected by any noise added to the sampler; the noise-free ODE cannot circumvent an infinite regression risk. $\square$

Theorem 7. (*Velocity blow-up*) Let ρ_0 be light-tailed (e.g. Gaussian) and ρ_1 regularly varying (Definition 1) with index α . Then the transport field (3.1.0.2) grows without bound in x : there is no constant L with $\|b^*(t, x)\| \leq L(1 + \|x\|)$ uniformly in t . The local Lipschitz hypothesis of Proposition 4, and the regularity assumption of Albergo and Vanden-Eijnden (2022), fail.

Proof. Work in one dimension; the radial argument in $\mathbb{R}^d$ is identical. Let F_t be the cumulative distribution function of ρ_t and recall that the probability-flow map $y_0 \mapsto y_t$ generated by b^* is the monotone (optimal-transport) map between marginals, so it transports the p -quantile of ρ_0 to the p -quantile of ρ_t ; equivalently the flow preserves the CDF level, $F_t(y_t) = F_0(y_0)$. Differentiating in t gives

$$b^*(t, y_t) = \dot{y}_t = -\frac{\partial_t F_t(y_t)}{\rho_t(y_t)} = \frac{d}{dt} F_t^{-1}(F_0(y_0)).$$

Consider the tail level $p \rightarrow 1$. If ρ_0 is Gaussian, its quantile grows as $F_0^{-1}(p) \sim \sqrt{2 \log \frac{1}{1-p}}$, whereas if ρ_1 is regularly varying with index α its quantile grows as $F_1^{-1}(p) \sim c(1-p)^{-1/\alpha}$. Along the interpolation the terminal quantile therefore exceeds the initial one by a factor $(1-p)^{-1/\alpha} / \sqrt{\log \frac{1}{1-p}} \rightarrow \infty$. Since the map must cover this ratio over $t \in [0, 1]$, its time derivative b^* evaluated at $y_t = F_t^{-1}(p)$ grows faster than any linear function of y_t as $p \rightarrow 1$, i.e. $\sup_t \|b^*(t, x)\| / (1 + \|x\|) \rightarrow \infty$ as $\|x\| \rightarrow \infty$. Thus no uniform Lipschitz bound exists, contradicting the hypothesis of Proposition 4 and the regularity assumption of Albergo and Vanden-Eijnden (2022). $\square$

Theorem 8. (*No transport equation for the conditional median*) Let the conditional law of $x_1 - x_0$ given x_t be asymmetric on a set of x_t of positive ρ_t -measure, so that $m(t, \cdot) \neq b^*(t, \cdot)$. Then:

1. the flow $\dot{y}_t = m(t, y_t)$ does not solve the continuity equation (3.1.0.3), and its time-1 law is in general not ρ_1 ;
2. no continuity equation is generated by m : unlike the mean, the conditional median does not satisfy a tower property $\text{med}[\cdot] = \mathbb{E}[\text{med}[\cdot \mid \mathcal{G}]]$, so the marginal drift cannot be assembled from conditional medians, and there is no law transported by m that reduces to ρ_1 at $t = 1$.

Proof. Write $\delta = x_1 - x_0$, let $b^*(t, x) = \mathbb{E}[\delta \mid x_t = x]$ and $m(t, x) = \text{med}[\delta \mid x_t = x]$, and assume $m(t, \cdot) \neq b^*(t, \cdot)$ on a set of positive ρ_t -measure.

(i) *The continuity equation is generated by the mean alone.* For every test function φ , the tower property gives

$$\frac{d}{dt} \mathbb{E} \varphi(x_t) = \mathbb{E} [\nabla \varphi(x_t) \cdot \delta] = \mathbb{E} [\nabla \varphi(x_t) \cdot b^*(t, x_t)],$$

the weak form of $\partial_t \rho_t + \nabla \cdot (b^* \rho_t) = 0$, which identifies b^* as the unique divergence-form field generating ρ_t . Any $m \neq b^*$ on a set of positive measure violates it for some φ , so the flow $\dot{y}_t = m(t, y_t)$ does not satisfy (3.1.0.3) and its time-1 law is not ρ_1 .

(ii) *No continuity equation is generated by the median.* The step above used linearity of $\mathbb{E}[\cdot \mid x_t]$ and its tower property. The median possesses neither. Explicitly, let $\mathcal{G} = \sigma(G)$ with $G \in \{0, 1\}$, $\mathbb{P}(G = 1) = 3/5$ and $v \mid G=0 \sim \mathcal{N}(0, \sigma^2)$, $v \mid G=1 \sim \mathcal{N}(a, \sigma^2)$ with $a \gg \sigma$; then $\text{med}[v \mid \mathcal{G}] \in \{0, a\}$, so $\mathbb{E}[\text{med}[v \mid \mathcal{G}]] = 3a/5$, whereas the mixture median lies inside the upper component, $\text{med}[v] = a - O(\sigma)$, giving $\text{med}[v] \neq \mathbb{E}[\text{med}[v \mid \mathcal{G}]]$. Because the flux appearing in the weak formulation can only be assembled from conditional quantities through such a tower step, and the median admits none, there is no divergence-form field built from m that reproduces $\partial_t \rho_t$. Hence, no continuity equation in m exists.

(iii) *The ghost law.* The flow $\dot{y}_t = m(t, y_t)$ still transports *some* family of laws; call it μ_t , with $\mu_0 = \rho_0$. By construction μ_t solves $\partial_t \mu_t + \nabla \cdot (m \mu_t) = 0$, whose generator is m rather than b^* . Subtracting the true continuity equation (3.1.0.3) for ρ_t ,

$$\partial_t (\mu_t - \rho_t) + \nabla \cdot (m \mu_t - b^* \rho_t) = 0,$$

so the discrepancy is driven by $\nabla \cdot ((m - b^*) \rho_t)$ to first order and accumulates monotonically wherever $m - b^*$ does not average to zero. Since $m - b^* \neq 0$ on a set of positive measure by hypothesis, and in the asymmetric conditional laws arising between separated modes $m - b^*$ does not change sign, $\mu_1 \neq \rho_1$; the flow converges to the ghost law rather than the target. $\square$

A.2 Proofs for Chapter 4

Throughout, p_{wm} denotes the working-model marginal, $\Lambda = p_{\text{data}}/p_{\text{wm}}$ the lift likelihood ratio, and $m_v = \mu + \beta v$. We restate each result before proving it.

A.2.1 The generalised inverse Gaussian law

Definition 34 (Generalised inverse Gaussian law). A random variable $v > 0$ follows the *generalised inverse Gaussian* law $\text{GIG}(\lambda, \chi, \psi)$ if it has density

$$p(v) = \frac{(\psi/\chi)^{\lambda/2}}{2K_\lambda(\sqrt{\chi\psi})} v^{\lambda-1} \exp\left\{-\frac{1}{2}(\chi v^{-1} + \psi v)\right\}, \quad v > 0, \quad (\text{A.2.1.1})$$

where K_λ is the modified Bessel function of the second kind and the parameters satisfy $\chi > 0, \psi \geq 0$ if $\lambda < 0$; $\chi > 0, \psi > 0$ if $\lambda = 0$; and $\chi \geq 0, \psi > 0$ if $\lambda > 0$. The choice $(\lambda, \chi, \psi) = (-v/2, v, 0)$ gives the inverse-gamma law $\text{InvGamma}(v/2, v/2)$; $\chi = 0, \lambda > 0$ gives the gamma law; and $\lambda = -1/2$ gives the inverse Gaussian law.

A.2.2 The posterior lift

Proposition 11 (Exactness of the lift). *For any working model (μ, β, p) with $p_{\text{wm}} > 0$ on the support of p_{data} , the x -marginal of the lifted joint is the data law: $\int \tilde{p}(x_0, v) dv = p_{\text{data}}(x_0)$. Consequently, learning the lifted conditionals $\tilde{p}(\cdot | v)$ and integrating v out targets p_{data} with no bias arising from the lift.*

Proof. By Definition 9 the map $v \mapsto q(v | x_0)$ in (4.1.1.2) is a posterior density and so integrates to one, since $\int_0^\infty \mathcal{N}(x_0; m_v, vI_d) p(v) dv = p_{\text{wm}}(x_0)$. Hence $\int_0^\infty \tilde{p}(x_0, v) dv = p_{\text{data}}(x_0)$ by (4.1.1.3), irrespective of (μ, β, p) . $\square$

Proposition 12 (Conjugacy of the GIG lift). *Let the working model of Definition 9 have mixing law $p = \text{GIG}(\lambda, \chi, \psi)$. Then the lift kernel (4.1.1.2) is again generalised inverse Gaussian,*

$$q(v | x_0) = \text{GIG}\left(\lambda - \frac{d}{2}, \chi + \|x_0 - \mu\|^2, \psi + \|\beta\|^2\right). \quad (\text{4.1.2.1})$$

In particular the Student- t lift ($\lambda = -v/2, \chi = v, \psi = 0, \beta = 0$) is inverse-gamma throughout: $q(v | x_0) = \text{InvGamma}(\frac{v+d}{2}, \frac{v+\|x_0-\mu\|^2}{2})$. For a product working model applied per coordinate, (4.1.2.1) holds coordinatewise with $\lambda - \frac{1}{2}, \chi + (x_{0,j} - \mu_j)^2$ and $\psi + \beta_j^2$.

Proof. Dropping factors independent of v and expanding $\|x_0 - \mu - \beta v\|^2 = \|x_0 - \mu\|^2 - 2v\langle x_0 - \mu, \beta \rangle + v^2\|\beta\|^2$, the v -free middle term is absorbed by normalisation and

$$q(v | x_0) \propto v^{(\lambda-d/2)-1} \exp\left\{-\frac{1}{2}[(\chi + \|x_0 - \mu\|^2)v^{-1} + (\psi + \|\beta\|^2)v]\right\},$$

which by Definition 34 is the kernel of $\text{GIG}(\lambda - \frac{d}{2}, \chi + \|x_0 - \mu\|^2, \psi + \|\beta\|^2)$. For the Student- t lift the identity $\text{GIG}(-a, \chi, 0) = \text{InvGamma}(a, \chi/2)$ gives the stated form, and the coordinatewise statement is the scalar computation applied per dimension. $\square$

Proposition 14 (Conditional light tails). *The lifted conditional factorises as*

$$\tilde{p}(x_0 | v) \propto \mathcal{N}(x_0; m_v, vI_d) \Lambda(x_0), \quad \Lambda = p_{\text{data}}/p_{\text{wm}}. \quad (4.1.2.2)$$

Under Assumption 13, for every $v > 0$ there is a constant $C(v) < \infty$ with $\tilde{p}(x_0 | v) \leq C(v) \exp\{-\|x_0 - m_v\|^2/(4v)\}$; in particular $\tilde{p}(\cdot | v)$ is sub-Gaussian and has finite moments of all orders, however heavy the tails of p_{data} may be.

Proof. Factorisation (4.1.2.2) is immediate from $\tilde{p}(x_0 | v) = \tilde{p}(x_0, v)/\tilde{p}(v)$ with $\tilde{p}(x_0, v) = p_{\text{data}}(x_0)q(v | x_0)$ and (4.1.1.2):

$$\tilde{p}(x_0 | v) = \frac{p_{\text{data}}(x_0) \mathcal{N}(x_0; m_v, vI_d) p(v)/p_{\text{wm}}(x_0)}{\tilde{p}(v)} \propto \mathcal{N}(x_0; m_v, vI_d) \Lambda(x_0), \quad \Lambda = p_{\text{data}}/p_{\text{wm}},$$

the v -dependent constant $p(v)/\tilde{p}(v)$ being independent of x_0 . For the bound, Assumption 13 gives $\Lambda(x_0) \leq C(1 + \|x_0\|)^k$, so

$$\tilde{p}(x_0 | v) \leq C'(v) (1 + \|x_0\|)^k \exp\left\{-\frac{\|x_0 - m_v\|^2}{2v}\right\}.$$

Since $(1 + \|x_0\|)^k \exp\{-\|x_0 - m_v\|^2/(2v)\} \leq C''(v) \exp\{-\|x_0 - m_v\|^2/(4v)\}$ for a finite $C''(v)$, the stated sub-Gaussian bound holds with $C(v) = C'(v)C''(v)$. Finiteness of all moments follows. $\square$

Remark 35 (The assumption is two-sided under a skewed working model). When $\beta \neq 0$ and $\psi = 0$, the working marginal p_{wm} is a skew- t , whose two tails have different indices: polynomial with index $v/2$ in the direction β , where the location term βv dominates, and exponentially light in the opposite direction, where that same term suppresses the large excursions of $\sqrt{v}\epsilon$. Assumption 13 then constrains p_{data} on each side separately, and is easiest to violate on the lighter side, where a symmetric heavy-tailed target may exceed the working model.

A.2.3 Tail inheritance

The following proofs concern the tail of the generated law. The symmetric and skew lifts are treated separately, since their exponents differ at two places that cancel. Recall from Definition 1 that a positive function $\bar{F}$ is regularly varying with index $-\kappa$, written $\bar{F} \in \text{RV}_{-\kappa}$, if $\bar{F}(u) = u^{-\kappa}L(u)$ with L slowly varying.

Theorem 16 (Tail inheritance). *Let $\hat{p}$ be the law from which the latent scale is drawn at generation time, and suppose the generated sample admits the representation $x = \mu + \beta v +$*

$\sqrt{v}y$ with $v \sim \hat{p}$ and, conditionally on v , $\mathbb{E}[\|y\|^{2\kappa+\delta} \mid v] \leq M < \infty$ for some $\delta > 0$ (guaranteed by Proposition 14). Suppose $\hat{p}$ is regularly varying (Definition 1) at infinity with index $-\kappa$, $\kappa > 0$. Then the tail of the generated law is determined by whichever of the two v -dependent terms dominates.

1. Symmetric case, $\beta = 0$. The scale term governs, and

$$\mathbb{P}(\|x\| > u) \sim \mathbb{E}[\|y\|^{2\kappa}] \mathbb{P}(\sqrt{v} > u), \quad u \rightarrow \infty, \quad (4.1.3.1)$$

so that $\|x\|$ is regularly varying with tail index 2κ .

2. Skewed case, $\beta \neq 0$. The location term governs, and

$$\mathbb{P}(\|x\| > u) \sim \mathbb{P}(\|\beta\|v > u) = \|\beta\|^\kappa \mathbb{P}(v > u) (1 + o(1)), \quad u \rightarrow \infty, \quad (4.1.3.2)$$

so that $\|x\|$ is regularly varying with tail index κ .

In both cases the tail index of the generated law is a functional of $\hat{p}$ alone, but the functional differs: it is 2κ under a symmetric working model and κ under a skewed one.

Proof. Recall $x = \mu + \beta v + \sqrt{v}y$ with $v \sim \hat{p} \in \text{RV}_{-\kappa}$ and, conditionally on v , $\mathbb{E}[\|y\|^{2\kappa+\delta} \mid v] \leq M < \infty$ by Proposition 14.

Symmetric case ($\beta = 0$). Then $x - \mu = \sqrt{v}y$. Since $v \in \text{RV}_{-\kappa}$,

$$\mathbb{P}(\sqrt{v} > u) = \mathbb{P}(v > u^2) = u^{-2\kappa} L(u^2), \quad (A.2.3.1)$$

and $u \mapsto L(u^2)$ is slowly varying, so $\sqrt{v} \in \text{RV}_{-2\kappa}$. The factor $\|y\|$ has a moment of order $2\kappa + \delta$ strictly exceeding the index 2κ , so Breiman's lemma (Breiman, 1965) applies to the product $\sqrt{v}\|y\|$ and gives

$$\mathbb{P}(\|x - \mu\| > u) \sim \mathbb{E}[\|y\|^{2\kappa}] \mathbb{P}(\sqrt{v} > u), \quad u \rightarrow \infty. \quad (A.2.3.2)$$

The fixed shift μ does not affect the tail, so $\|x\| \in \text{RV}_{-2\kappa}$.

Note that the two factors are not independent, since y is distributed conditionally on v through (4.3.2.2); the uniform bound $\mathbb{E}[\|y\|^{2\kappa+\delta} \mid v] \leq M$ is what allows the dependent form of the lemma, and the tail constant is then $\lim_{v \rightarrow \infty} \mathbb{E}[\|y\|^{2\kappa} \mid v]$ where that limit exists.

Skew case ($\beta \neq 0$). Write $b = \beta/\|\beta\|$. By the triangle inequality,

$$\|\beta\|v - \|\mu\| - \sqrt{v}\|y\| \leq \|x\| \leq \|\mu\| + \|\beta\|v + \sqrt{v}\|y\|. \quad (A.2.3.3)$$

The middle term of each bound lies in $\text{RV}_{-\kappa}$, since $\|\beta\|v$ is a positive multiple of v . The last term lies in $\text{RV}_{-2\kappa}$ by the symmetric case, and $2\kappa > \kappa$, so for any $\varepsilon \in (0, 1)$,

$$\mathbb{P}(\sqrt{v}\|y\| > \varepsilon u) = o(\mathbb{P}(v > u)), \quad u \rightarrow \infty, \quad (\text{A.2.3.4})$$

and the same holds for the constant $\|\mu\|$. Applying this to both sides of (A.2.3.3) and letting $\varepsilon \downarrow 0$,

$$\mathbb{P}(\|x\| > u) \sim \mathbb{P}(\|\beta\|v > u) = \|\beta\|^\kappa \mathbb{P}(v > u) (1 + o(1)), \quad (\text{A.2.3.5})$$

so $\|x\| \in \text{RV}_{-\kappa}$. The same argument applied to $\langle x, b \rangle$ shows the tail is carried by the direction β . The skew lift therefore produces a strictly heavier tail than the symmetric one at equal κ , index κ against 2κ . $\square$

The two cases are what make Corollary 18 hold under both working models. Essentially, the exponent in the data-space representation and the rate at which the posterior concentrates both change, and they change in compensating directions.

Corollary 18 (The lift is tail-adaptive). *Suppose p_{data} has regularly varying tails of index $\alpha > 0$ and let the working model have $\psi = 0$ and mixing $\text{GIG}(-v/2, v, 0) = \text{InvGamma}(v/2, v/2)$, for any working $v > 0$. Generation with $\hat{p} = \tilde{p}$ produces samples with tail index α in both of the following cases, independently of the working v .*

1. Student- t lift, $\beta = 0$. The posterior scale parameter is $\chi' = v + \|x_0 - \mu\|^2$ and $\psi' = 0$, so $\mathbb{E}[v \mid x_0] = \chi'/(v + d - 2) \asymp \|x_0 - \mu\|^2$. Provided $\alpha < v + d$, the normalised posterior has a moment of order $\alpha/2 + \delta'$ and $\tilde{p} \in \text{RV}_{-\alpha/2}$, so Theorem 16(i) with $\kappa = \alpha/2$ gives index $2\kappa = \alpha$. The condition is not restrictive, as it fails only for a working model heavier-tailed than the data in low dimension.
2. Skew- t lift, $\beta \neq 0$. The posterior parameters are $\chi' = v + \|x_0 - \mu\|^2$ and $\psi' = \|\beta\|^2 > 0$, so as $\chi'\psi' \rightarrow \infty$ the GIG mean satisfies $\mathbb{E}[v \mid x_0] \sim \sqrt{\chi'/\psi'} \asymp \|x_0 - \mu\|/\|\beta\|$. Hence $\tilde{p} \in \text{RV}_{-\alpha}$, and Theorem 16(ii) with $\kappa = \alpha$ gives index $\kappa = \alpha$.

Proof. Write $r = \|x_0 - \mu\|$ and take the working model of Corollary 18, so that $\psi = 0$ and the prior is $\text{InvGamma}(v/2, v/2)$ for an arbitrary working $v > 0$. By Proposition 12 the posterior is $\text{GIG}(\lambda', \chi', \psi')$ with

$$\lambda' = -\frac{v+d}{2}, \quad \chi' = v + r^2, \quad \psi' = \|\beta\|^2. \quad (\text{A.2.3.6})$$

The two cases differ only in whether ψ' vanishes.

We use throughout the following hypothesis, which isolates the one step requiring care.

(H) Let $S(r)$ denote the scale defined in each case below and $w := v/S(r)$. The conditional law of w given x_0 converges as $r \rightarrow \infty$ to a limit law w_∞ , and the convergence is uniform enough that $\mathbb{P}(v > u) = \mathbb{E}_{x_0}[\bar{F}_{w_\infty}(u/S(r))](1 + o(1))$.

Remark 36 (Standing hypothesis, and what is not claimed). The argument rests on (H), which permits Breiman's lemma to be applied to the pair $(S(r), w)$ although these are dependent. Establishing (H) requires the convergence $w | x_0 \rightarrow w_\infty$ to be uniform in a sense we do not verify here. We claim the index only, and not the sharpness of the slowly varying factor. Both exponents are confirmed numerically in Section 6.5, on the symmetric and the skew branch.

Case (i): symmetric lift, $\beta = 0$, so $\psi' = 0$. The posterior reduces to $\text{InvGamma}(a, b)$ with $a = (v + d)/2$ and $b = (v + r^2)/2$, so $\mathbb{E}[v | x_0] = (v + r^2)/(v + d - 2)$ and the natural scale is $S(r) = r^2$. Since $v/b \sim \text{InvGamma}(a, 1)$ exactly, and $b/r^2 \rightarrow \frac{1}{2}$,

$$w = \frac{v}{r^2} = \frac{b}{r^2} \cdot \frac{v}{b} \xrightarrow{d} w_\infty = \frac{1}{2} U, \quad U \sim \text{InvGamma}\left(\frac{v+d}{2}, 1\right). \quad (\text{A.2.3.7})$$

The moment $\mathbb{E}[U^p]$ is finite exactly for $p < a$, and

$$\mathbb{E}[w^p | x_0] = \frac{b^p}{r^{2p}} \frac{\Gamma(a - p)}{\Gamma(a)} \longrightarrow 2^{-p} \frac{\Gamma(a - p)}{\Gamma(a)}, \quad (\text{A.2.3.8})$$

so the family is uniformly bounded in r for any $p < a$. Breiman's lemma requires a moment of order $\alpha/2 + \delta'$ for some $\delta' > 0$, which by (A.2.3.8) holds precisely when

$$\alpha < v + d. \quad (\text{A.2.3.9})$$

Under (A.2.3.9) and (H), since $p_{\text{data}} \in \text{RV}_{-\alpha}$ gives $r^2 \in \text{RV}_{-\alpha/2}$, Breiman applied to $v = r^2 w$ yields $\tilde{p} \in \text{RV}_{-\alpha/2}$ with tail constant $\mathbb{E}[w_\infty^{\alpha/2}]$. The exponent does not involve v . Applying Theorem 16(i) with $\kappa = \alpha/2$ gives a generated tail index $2\kappa = \alpha$.

Case (ii): skew lift, $\beta \neq 0$, so $\psi' = \|\beta\|^2 > 0$. Set $\omega = \sqrt{\chi' \psi'} = \|\beta\| \sqrt{v + r^2}$, which tends to infinity with r . For $\chi', \psi' > 0$ the GIG has moments of every order,

$$\mathbb{E}[v^p | x_0] = \left(\frac{\chi'}{\psi'}\right)^{p/2} \frac{K_{\lambda'+p}(\omega)}{K_{\lambda'}(\omega)}, \quad p \in \mathbb{R}, \quad (\text{A.2.3.10})$$

and $K_{\lambda+p}(\omega)/K_{\lambda}(\omega) \rightarrow 1$ as $\omega \rightarrow \infty$ by the uniform asymptotic $K_{\lambda}(\omega) \sim \sqrt{\pi/2\omega} e^{-\omega}$. Taking $p = 1$ in (A.2.3.10),

$$\mathbb{E}[v | x_0] \sim \sqrt{\chi'/\psi'} = \frac{\sqrt{v + r^2}}{\|\beta\|} \sim \frac{r}{\|\beta\|}, \quad (\text{A.2.3.11})$$

so the natural scale is $S(r) = r/\|\beta\|$, linear in the radius rather than quadratic. Moreover, taking $p = 1, 2$ in (A.2.3.10) and expanding $K_{\lambda+1}(\omega)/K_{\lambda}(\omega) = 1 + (2\lambda + 1)/2\omega + O(\omega^{-2})$ gives

$$\frac{\text{Var}(v \mid x_0)}{\mathbb{E}[v \mid x_0]^2} = O(\omega^{-1}) = O(r^{-1}), \quad (\text{A.2.3.12})$$

so the posterior concentrates relatively and $w = v/S(r) \rightarrow 1$ in probability, that is $w_{\infty} = \delta_1$. Its moments are uniformly bounded in r : by (A.2.3.10),

$$\mathbb{E}[w^p \mid x_0] = \left(\frac{\|\beta\|}{r}\right)^p \left(\frac{v + r^2}{\|\beta\|^2}\right)^{p/2} \frac{K_{\lambda'+p}(\omega)}{K_{\lambda'}(\omega)} \rightarrow 1 \quad \text{for every } p \in \mathbb{R}. \quad (\text{A.2.3.13})$$

No analogue of (A.2.3.9) is therefore needed. Under (H), since $r \in \text{RV}_{-\alpha}$, Breiman applied to $v = S(r)w$ yields $\tilde{p} \in \text{RV}_{-\alpha}$, with tail constant $\|\beta\|^{\alpha} \mathbb{E}[w_{\infty}^{\alpha}] = \|\beta\|^{\alpha}$. Again the exponent does not involve v . Applying Theorem 16(ii) with $\kappa = \alpha$ gives a generated tail index $\kappa = \alpha$.

In both cases the generated law is regularly varying with index α , independently of the working v . $\square$

Remark 37 (What the two cases trade against each other). The symmetric posterior does not concentrate relatively: it is $\text{InvGamma}(a, b)$ with $a = (v + d)/2$, so for $v + d > 4$ its coefficient of variation is exactly $(a - 2)^{-1/2} = \sqrt{2/(v + d - 4)}$, independent of the radius, and this is what governs how far a finite latent pool sits from the asymptotic prediction. For $v + d \leq 4$ the posterior has no variance at all, which is the regime reported in Section 6.6.1. The skew posterior does concentrate, at rate $r^{-1/2}$. The compensating difference is in the exponent: the symmetric lift concentrates at scale r^2 and enters x through $\sqrt{v}$, the skew lift at scale r and enters through βv . Either way the generated index is α , but the intermediate index of $\tilde{p}$ is $\alpha/2$ in the first case and α in the second, which is the quantity to compare against when diagnosing a shortfall empirically (Section 6.5).

A.2.4 Conditional diffusion and objective

Proposition 21 (Scale invariance of the conditional SNR). *Define the conditional signal-to-noise ratio of (4.2.1.2) as $\text{SNR}(t \mid v) := s(t)^2 \text{trCov}(x_0 \mid v) / (d v \sigma(t)^2)$. Under the working model, $\text{Cov}(x_0 \mid v) = v I_d$ and hence*

$$\text{SNR}(t \mid v) = \frac{s(t)^2}{\sigma(t)^2}, \quad (4.2.1.4)$$

which is independent of v . In particular, relative to the conditional scale, the noise level reached at any given t is the same for every latent draw. The computation is under the

working model, which is what the $\sqrt{v}$ scaling is calibrated to; away from the exact-NMVM point of Corollary 30 the lifted conditional has covariance vI_d only up to the tilt Λ , and the v -freeness is correspondingly approximate.

Proof. Under the working model $\text{Cov}(x_0 | v) = vI_d$, so the signal power is $s(t)^2 \text{tr Cov}(x_0 | v) = s(t)^2 vd$ and the noise power is $d v \sigma(t)^2$. Their ratio is $s(t)^2 vd / (d v \sigma(t)^2) = s(t)^2 / \sigma(t)^2$, free of v . $\square$

Theorem 24 (Conditional time reversal). *Under Assumption 23, (2.1.2.1) holds for each fixed v : the process solving, backwards from $t = T$ to $t = 0$,*

$$dx_t = \left[f(t)x_t - v g(t)^2 \nabla_x \log \tilde{p}_t(x_t | v) \right] dt + \sqrt{v} g(t) d\bar{B}_t, \quad x_T \sim \tilde{p}_T(\cdot | v), \quad (4.2.2.1)$$

has law $\tilde{p}_t(\cdot | v)$ at every t , as does the probability-flow ODE

$$\frac{dx_t}{dt} = f(t)x_t - \frac{v g(t)^2}{2} \nabla_x \log \tilde{p}_t(x_t | v). \quad (4.2.2.2)$$

Conditioning is legitimate because v is static: the generator of (x_t, v) acts on the x -coordinate alone, so the reversal of the pair is the v -wise reversal of x .

Proof. Fix v . By Proposition 14 and Lemma 20 the conditional marginals $\tilde{p}_t(\cdot | v)$ are smooth, strictly positive and sub-Gaussian, so Assumption 23 holds and Anderson's theorem (Anderson, 1982) applies to the linear SDE (4.2.1.1), yielding the reverse SDE (4.2.2.1). Because v is static (i.e. the generator of (x_t, v) differentiates in x only), the time reversal of the joint process is the reversal of the x -dynamics at each v , so no v -reversal term appears. The probability-flow ODE (4.2.2.2) is the deterministic process with the same Fokker–Planck marginals, obtained by removing the diffusion term and halving the score drift (Song et al., 2021b). $\square$

Theorem 26 (Conditional DSM equivalence). *Under Assumption 23, for every weighting ω we have $\mathcal{L}_{\text{DSM}}(\theta) = \mathcal{L}_{\text{SM}}(\theta) + C$ with $C \geq 0$ independent of θ , and both risks are minimised by the true conditional score $s^*(x, t, v) = \nabla_x \log \tilde{p}_t(x | v)$. This is the identity behind (2.1.3.1) carried out at fixed v , which is legitimate because v is not integrated over inside the conditional expectation.*

Proof. Fix v . This is the argument behind (2.1.3.1) carried out at fixed v , which is legitimate because v is not integrated over inside the conditional expectation. Expanding (4.3.1.2), the only θ -dependent cross term uses $p_{0t} \nabla_{x_t} \log p_{0t} = \nabla_{x_t} p_{0t}$ and Fubini,

$$\mathbb{E}_{x_0|v} [p_{0t}(x_t | x_0, v) \nabla_{x_t} \log p_{0t}] = \nabla_{x_t} \tilde{p}_t(x_t | v),$$

so it coincides with the cross term of $\mathcal{L}_{\text{SM}}$, and the remaining terms differ by a θ -free constant $C \geq 0$. Taking expectations over $v \sim \tilde{p}(v)$ preserves the identity, and both risks are non-negative quadratics minimised at the true conditional score. $\square$

Proposition 27 (Regression target and its second moment). *The denoising target of (4.3.1.2) is*

$$\nabla_{x_t} \log p_{0t}(x_t | x_0, v) = -\frac{x_t - s(t)x_0}{v\sigma(t)^2} = -\frac{\varepsilon}{\sqrt{v}\sigma(t)}, \quad (4.3.1.3)$$

and its conditional second moment is

$$\mathbb{E}\left[\left\|\nabla_{x_t} \log p_{0t}(x_t | x_0, v)\right\|^2 \middle| v\right] = \frac{d}{v\sigma(t)^2}. \quad (4.3.1.4)$$

Proof. From Lemma 20, $\log p_{0t}(x_t | x_0, v) = -\|x_t - s(t)x_0\|^2/(2v\sigma(t)^2) + \text{const}$, so $\nabla_{x_t} \log p_{0t} = -(x_t - s(t)x_0)/(v\sigma(t)^2)$. Substituting $x_t - s(t)x_0 = \sigma(t)\sqrt{v}\varepsilon$ gives $-\varepsilon/(\sqrt{v}\sigma(t))$, proving (4.3.1.3). Its squared norm has conditional expectation $\mathbb{E}[\|\varepsilon\|^2]/(v\sigma(t)^2) = d/(v\sigma(t)^2)$, which is (4.3.1.4). $\square$

Corollary 38 (Finiteness under latent-uniform weighting). *Under a weighting $\omega = \omega(t)$ that does not depend on v , the risk (4.3.1.2) is finite if and only if*

$$\mathbb{E}_{\tilde{p}(v)}[v^{-1}] < \infty \quad \text{and} \quad \int_0^T \frac{\omega(t)}{\sigma(t)^2} dt < \infty. \quad (\text{A.2.4.1})$$

Were v drawn from the prior $\text{GIG}(\lambda, \chi, \psi)$, the first condition would hold if and only if $\chi > 0$ or $\chi = 0$ and $\lambda > 1$. Remark 39 shows the lift discharges it regardless.

Proof. By Proposition 27 the weighted second moment of the regression target is

$$\mathbb{E}\left[\omega(t)\left\|\nabla_{x_t} \log p_{0t}(x_t | x_0, v)\right\|^2\right] = \int_0^T \omega(t) \mathbb{E}_{\tilde{p}(v)}\left[\frac{d}{v\sigma(t)^2}\right] dt = d \mathbb{E}_{\tilde{p}(v)}[v^{-1}] \int_0^T \frac{\omega(t)}{\sigma(t)^2} dt.$$

Let s_θ satisfy $\mathbb{E}[\omega\|s_\theta\|^2] < \infty$. The inequality $\|u - w\|^2 \geq \frac{1}{2}\|w\|^2 - \|u\|^2$ gives $\mathcal{L}_{\text{DSM}}(\theta) \geq \frac{1}{2}\mathbb{E}[\omega\|\nabla \log p_{0t}\|^2] - \mathbb{E}[\omega\|s_\theta\|^2]$, so the risk is infinite whenever that moment is, while $\|u - w\|^2 \leq 2\|u\|^2 + 2\|w\|^2$ gives the converse. Finiteness therefore holds exactly when both factors above are finite. For $\text{GIG}(\lambda, \chi, \psi)$ mixing the density behaves as $v^{\lambda-1}e^{-\chi/(2v)}$ near 0, so $\mathbb{E}[v^{-1}]$ converges at the origin if and only if $\chi > 0$, or $\chi = 0$ with $\lambda > 1$. $\square$

Remark 39 (The posterior lift discharges the moment condition). The condition constrains the *lifted* latent marginal, not the working prior, and the lift improves it. By Proposition 12 the posterior parameter is $\chi' = \chi + \|x_0 - \mu\|^2$, strictly positive p_{data} -almost surely unless p_{data} has an atom at μ ; for the Student- t lift the conditional moment is closed-form, $\mathbb{E}[v^{-1} |$

$x_0] = (\nu + d)/(\nu + \|x_0 - \mu\|^2) \leq (\nu + d)/\nu$, so $\mathbb{E}_{\tilde{p}(\nu)}[\nu^{-1}]$ is finite uniformly in the data. Under any skewed working model the conclusion is stronger still: $\psi' = \|\beta\|^2 > 0$ gives a GIG posterior with moments of every order.

Theorem 28 (Scale-adapted weighting and noise prediction). *Adopt the weighting $\omega(t, \nu) = \nu\sigma(t)^2$ and reparameterise*

$$\varepsilon_\theta(x, t, \nu) := -\sqrt{\nu}\sigma(t)s_\theta(x, t, \nu). \quad (4.3.1.5)$$

Then the risk (4.3.1.2) equals the noise-prediction objective

$$\mathcal{L}(\theta) = \mathbb{E}_{t, (x_0, \nu) \sim \tilde{p}, \varepsilon} \left[\left\| \varepsilon_\theta(s(t)x_0 + \sigma(t)\sqrt{\nu}\varepsilon, t, \nu) - \varepsilon \right\|^2 \right], \quad (4.3.1.6)$$

whose regression target is standard normal. Consequently (4.3.1.6) has finite value and finite-variance targets for any mixing law, with no moment condition on $\tilde{p}(\nu)$; its optimal value $\mathbb{E}\|\varepsilon - \mathbb{E}[\varepsilon | x_t, \nu]\|^2 \leq d$ is uniformly bounded; and its minimiser is the conditional Tweedie denoiser

$$\varepsilon^*(x, t, \nu) = \mathbb{E}[\varepsilon | x_t = x, \nu] = -\sqrt{\nu}\sigma(t)\nabla_x \log \tilde{p}_t(x | \nu). \quad (4.3.1.7)$$

Proof. With $\omega(t, \nu) = \nu\sigma(t)^2$ and $\varepsilon_\theta = -\sqrt{\nu}\sigma(t)s_\theta$, note from (4.3.1.3) that the DSM target is $-\varepsilon/(\sqrt{\nu}\sigma(t))$, so

$$\omega \|s_\theta - \nabla \log p_{0t}\|^2 = \nu\sigma(t)^2 \left\| -\frac{\varepsilon_\theta}{\sqrt{\nu}\sigma(t)} + \frac{\varepsilon}{\sqrt{\nu}\sigma(t)} \right\|^2 = \|\varepsilon_\theta - \varepsilon\|^2,$$

pointwise. Taking expectations turns (4.3.1.2) into (4.3.1.6), whose target ε is standard normal. The optimum of a conditional-mean-squared objective is the conditional mean $\varepsilon^* = \mathbb{E}[\varepsilon | x_t, \nu]$, with value $\mathbb{E}\|\varepsilon - \mathbb{E}[\varepsilon | x_t, \nu]\|^2 \leq \mathbb{E}\|\varepsilon\|^2 = d$; Tweedie's formula for the Gaussian kernel of Lemma 20 identifies $\mathbb{E}[\varepsilon | x_t, \nu] = -\sqrt{\nu}\sigma(t)\nabla_x \log \tilde{p}_t(x | \nu)$. $\square$

Proposition 29 (Scale equivariance). *Fix ν and define the standardised process $y_t := (x_t - s(t)m_\nu)/\sqrt{\nu}$. Then y_t satisfies the ν -free SDE*

$$dy_t = f(t)y_t dt + g(t)dB_t, \quad (4.3.2.1)$$

started from the standardised lifted conditional

$$q_0(y | \nu) \propto \mathcal{N}(y; 0, I_d) \Lambda(m_\nu + \sqrt{\nu}y), \quad \Lambda = p_{\text{data}}/p_{\text{wm}}. \quad (4.3.2.2)$$

Writing q_t for its marginals, the conditional score is equivariant,

$$\nabla_x \log \tilde{p}_t(x | v) = \frac{1}{\sqrt{v}} \nabla_y \log q_t \left(\frac{x - s(t)m_v}{\sqrt{v}} \mid v \right), \quad (4.3.2.3)$$

and all dependence on v beyond the affine standardisation is carried by the initial condition (4.3.2.2) alone.

Proof. With $y_t = (x_t - s(t)m_v)/\sqrt{v}$ and $\dot{s} = fs$,

$$dy_t = \frac{dx_t - \dot{s}(t)m_v dt}{\sqrt{v}} = \frac{f(t)x_t dt + \sqrt{v}g(t)dB_t - f(t)s(t)m_v dt}{\sqrt{v}} = f(t) \frac{x_t - s(t)m_v}{\sqrt{v}} dt + g(t) dB_t,$$

which is (4.3.2.1), free of v in its coefficients. The law of y_0 is the pushforward of $\tilde{p}(x_0 | v) \propto \mathcal{N}(x_0; m_v, vI_d)\Lambda(x_0)$ under $x_0 \mapsto (x_0 - m_v)/\sqrt{v}$; the Gaussian factor becomes $\mathcal{N}(y; 0, I_d)$ and $\Lambda(x_0) = \Lambda(m_v + \sqrt{v}y)$, giving (4.3.2.2). The score identity (4.3.2.3) is the chain rule applied to $\tilde{p}_t(x | v) = v^{-d/2} q_t((x - s(t)m_v)/\sqrt{v} | v)$. $\square$

Corollary 30 (Exact-NMVM closed form). *If p_{data} coincides with the working-model marginal p_{wm} , then $\Lambda \equiv 1$ and (4.3.2.2) reduces to the v -free law $q_0 = \mathcal{N}(0, I_d)$; hence $q_t = \mathcal{N}(0, (s(t)^2 + \sigma(t)^2)I_d)$ and the optimal denoiser is independent of v in standardised coordinates. Under a variance-preserving schedule it is the linear map $\varepsilon^*(x, t, v) = \sqrt{1 - \bar{\alpha}_t} y_t$.*

Proof. If $p_{\text{data}} = p_{\text{wm}}$ then $\Lambda \equiv 1$, so (4.3.2.2) gives $q_0(y | v) = \mathcal{N}(y; 0, I_d)$, free of v . Propagating through the v -free SDE (4.3.2.1) yields $q_t = \mathcal{N}(0, (s(t)^2 + \sigma(t)^2)I_d)$; back-substituting (4.3.2.3) and the VP identities $s(t) = \sqrt{\bar{\alpha}_t}$, $s^2 + \sigma^2 = 1$ gives $\varepsilon^* = \sqrt{1 - \bar{\alpha}_t} y_t$. $\square$

A.2.5 Generation bound

Assumption 40 (Conditional Girsanov regularity). For each $v > 0$ the learned score $s_\theta(\cdot, \cdot, v)$ and the true conditional score $\nabla \log \tilde{p}_t(\cdot | v)$ are such that the Girsanov change of measure between the reverse SDE (4.2.2.1) driven by each is well defined; it suffices that $\int_0^T \mathbb{E}_{\tilde{p}_t(\cdot | v)} \|s_\theta - \nabla \log \tilde{p}_t(\cdot | v)\|^2 dt < \infty$ and that the corresponding Novikov condition holds.

Theorem 41 (Conditional generation bound). *Let $\hat{p}_0(\cdot | v)$ denote the law produced by the reverse SDE (4.2.2.1) with s_θ in place of the true conditional score, initialised at π_v . Under Assumptions 23 and 40, for each v ,*

$$\text{KL}(\tilde{p}_0(\cdot | v) \parallel \hat{p}_0(\cdot | v)) \leq \underbrace{\text{KL}(\tilde{p}_T(\cdot | v) \parallel \pi_v)}_{\text{prior mismatch}} + \frac{1}{2} \int_0^T v g(t)^2 \mathbb{E}_{\tilde{p}_t(\cdot | v)} \|s_\theta - \nabla \log \tilde{p}_t(\cdot | v)\|^2 dt. \quad (\text{A.2.5.1})$$

Proof. Fix v . By Proposition 14 and lemma 20 the conditional marginals are sub-Gaussian with smooth scores, so Assumption 40 holds and the standard KL bound for score-based samplers (Chen et al., 2023, Thm. 2) applies verbatim to the conditional reverse SDE (4.2.2.1), giving (4.3.3.1); the static latent enters only as the constant v in the diffusion coefficient. We do not reproduce the Girsanov argument, which is not original to this work. $\square$

Corollary 42 (Marginal generation bound). *Let $\hat{p}_0$ be the generated marginal obtained by drawing $v \sim \tilde{p}(v)$, sampling $x \sim \hat{p}_0(\cdot | v)$, and discarding v . Then, by exactness of the lift (Proposition 11) and convexity of total variation,*

$$\text{TV}(p_{\text{data}}, \hat{p}_0) \leq \mathbb{E}_{\tilde{p}(v)} \left[\text{TV}(\tilde{p}_0(\cdot | v), \hat{p}_0(\cdot | v)) \right] \leq \mathbb{E}_{\tilde{p}(v)} \sqrt{\frac{1}{2} \text{KL}(\tilde{p}_0(\cdot | v) \| \hat{p}_0(\cdot | v))}. \quad (\text{A.2.5.2})$$

The marginal generation error is thus the $\tilde{p}(v)$ -average of per- v conditional errors, each of which falls under classical finite-moment convergence theory that the heavy-tailed marginal itself violates.

Proof. By Proposition 11 the $\tilde{p}(v)$ -mixture of $\tilde{p}_0(\cdot | v)$ is p_{data} , and the same mixture of $\hat{p}_0(\cdot | v)$ is $\hat{p}_0$. Convexity of total variation gives $\text{TV}(p_{\text{data}}, \hat{p}_0) \leq \mathbb{E}_{\tilde{p}(v)} \text{TV}(\tilde{p}_0(\cdot | v), \hat{p}_0(\cdot | v))$; Pinsker's inequality bounds each conditional TV by $\sqrt{\frac{1}{2} \text{KL}}$, and Jensen moves the expectation inside the square root if desired. $\square$

A.3 Proofs for Chapter 5

Lemma 32 (Discriminating lemma). *Let p_{data} be arbitrary and consider the template (5.1.0.1).*

1. *If $u \perp\!\!\!\perp x_0$, then $q(u | x_0) = p(u)$ and the conditional noise scale is constant in x_0 wherever it exists. This covers t -EDM, t -Flow, DLPM and HTFM. For the Student- t instances it equals $\mathbb{E}[w] = v/(v-2)$, and hence requires $v > 2$.*
2. *Under the posterior lift (5.1.0.2) with $\text{GIG}(\lambda, \chi, \psi)$ mixing, the conditional law of v is the posterior (4.1.2.1) and depends on x_0 through $\|x_0 - \mu\|^2$. For the Student- t lift,*

$$\zeta_{\text{lift}}(x_0) = \mathbb{E}[v | x_0] = \frac{v + \|x_0 - \mu\|^2}{v + d - 2}, \quad v + d > 2, \quad (5.4.0.1)$$

which is strictly increasing and unbounded in $\|x_0 - \mu\|$.

3. *Hence the two families of forward process have different conditional covariance structures, constant against affine in $\|x_0 - \mu\|^2$, and are distinct stochastic processes.*

Proof. Part (i) is immediate from the independence $w \perp\!\!\!\perp x_0$ built into the t-EDM and DLPM instances of (5.1.0.1); the value $\mathbb{E}[w] = v/(v-2)$ is the mean of $\text{InvGamma}(v/2, v/2)$. For (ii), Proposition 12 with $\beta = 0$, $\lambda = -v/2$, $\chi = v$, $\psi = 0$ gives $v | x_0 \sim \text{InvGamma}(\frac{v+d}{2}, \frac{v+\|x_0-\mu\|^2}{2})$, whose mean is $b/(a-1)$ with $a = (v+d)/2$, $b = (v+\|x_0-\mu\|^2)/2$, giving (5.4.0.1). Part (iii) follows since a constant function and a strictly increasing unbounded one cannot coincide. $\square$

Remark 43 (Existence of the conditional scale). The scale $v/(v-2)$ requires $v > 2$, the condition that also appears in the $v/(v-2)$ factors of the t-EDM preconditioning. For $v < 2$ the scale does not exist for either instance, so Lemma 32 does not apply; the difference is that t-EDM's preconditioning uses that moment and breaks, whereas t-Flow does not use it. The lifted scale (5.4.0.1) requires only $v+d > 2$, hence holds for every $v > 0$ whenever $d \geq 2$: by Proposition 12 conditioning adds $d/2$ to the shape of the posterior, which is thus better behaved than the prior.

A.4 Proofs for Chapter 6

Proposition 33 (Non-convergence of sample moment ratios). *Let $X_1, \dots, X_n$ be i.i.d. with $|X| \in \text{RV}_{-\alpha}$ and let $g_1^{(n)}, g_2^{(n)}$ be the sample skewness and kurtosis. Then $g_1^{(n)} \rightarrow \infty$ in probability if $\alpha < 3$ and $g_2^{(n)} \rightarrow \infty$ if $\alpha < 4$, so the ratios formed from two such samples have no deterministic limit.*

Proof. Write $m_k = n^{-1} \sum_i (X_i - \bar{X})^k$, so $g_1^{(n)} = m_3/m_2^{3/2}$ and $g_2^{(n)} = m_4/m_2^2$. Under $\mathbb{P}(|X| > u) = u^{-\alpha} L(u)$ the variable $|X|^k$ is regularly varying with index α/k , so $\mathbb{E}|X|^k < \infty$ exactly when $k < \alpha$. When $k \geq \alpha$ the generalised central limit theorem (Samorodnitsky and Taqqu, 1994) gives $n^{-k/\alpha} \sum_i |X_i - \bar{X}|^k \rightsquigarrow S_k$ for a positive (α/k) -stable S_k , so m_k has exact order $n^{k/\alpha-1}$ with a non-degenerate random prefactor; centring is negligible, since $\bar{X} = o_p(n^{1/\alpha})$ against a maximal summand of order $n^{k/\alpha}$.

For $\alpha < 3$ take $k = 3$: if $\alpha > 2$ then $m_2 \rightarrow \sigma^2$ while $m_3 \rightarrow \infty$, and if $\alpha \leq 2$ then m_3 is of order $n^{3/\alpha-1}$ against $m_2^{3/2}$ of order $n^{3/\alpha-3/2}$, so $g_1^{(n)} \asymp n^{1/2}$. For $\alpha < 4$ take $k = 4$: if $\alpha > 2$ then $m_4 \rightarrow \infty$ with m_2 bounded, and if $\alpha \leq 2$ then $g_2^{(n)} \asymp n$. Both diverge in probability. The ratios are measurable functions of (g_1, g_2) on two independent samples, each diverging at a non-degenerate random rate, so they have no deterministic limit and their ordering across models is not consistent as $n \rightarrow \infty$. $\square$

Appendix B

Experimental harness

This appendix collects the numerical detail referenced from Chapter 6. The metric suite, floor computation and sweep protocol are implemented as a reusable harness; every number below is derived from the recorded run files rather than transcribed from a log.

B.1 Targets, preprocessing and training configuration

Target	Construction	d	ν	β	Train	Gen.
<i>Vectors</i>						
Grid mixture	nine NMVM components, 3×3 grid	2	2.5, 1.5	0	20000	20000
Skewed vector	unimodal NMVM	8, 32	2.5	0.5	20000	20000
Misspecification	unimodal NMVM	8, 32	5	1	20000	20000
<i>Fields</i>						
Scalar ν	$x(s) = \sqrt{\nu}z(s)$, one ν per field	256	2.5, 1.5	0	20000	20000
Local scale	$\nu(s)$ smooth, corr. length $\ell_\nu \in \{2, 3, 4\}$	256	2.5, 1.5	0	20000	20000
Gaussian copula	Student- t margins, asymptotically indep.	256	2.5, 1.5	0	20000	20000
Skewed copula	as above, skew- t margins, $\gamma = 2$	256	2.5	—	20000	20000
HRRR w20	analysis crops, 2019–2020 train, 2021 test	1024	3.53	—	15,000	7000
HRRR VIL	as above, crops with maximum above 1.0	1024	1.14	—	14,722	6921

Table B.1 The targets. Synthetic tail indices are exact by construction; the HRRR figures are pooled-pixel Hill estimates. The local-scale field is normalised by the exact kernel norm rather than per sample, which would make ν recoverable from one field and turn the lift into an oracle. The VIL filter provably removes no tail mass.

Target class	Scaler
$\alpha \geq 2$	z-score on the training split
$\alpha < 2$	(median + 1.4826×MAD) or (global z-score)
Fields	one global scalar, not per pixel
Multimodal	one global scale
VIL	high quantile of positive training pixels, no centring

Table B.2 Normalisation by target class, fitted on the training split and identical across models within an experiment. The inverse is exact and every metric runs in raw coordinates.

Target group	Network	Parameters	Lift overhead
Vector	three-layer perceptron, width 256, GELU	66.8k	0.2%
Fields, 16×16	DDPM-style U-Net, group normalisation, adaptive scale-and-shift conditioning	1.65M	1.0%
HRRR, 32×32	as above, with one extra layer and attention at resolution 16	4.48M	$\leq 1.0\%$

Table B.3 Architectures. One family per target type, identical across models within an experiment; the only model-specific difference is the conditioning input the lift requires. Referenced from Section 6.3.

Target group	Epochs	Batch	Samples	Seeds	Sampler
Vector, $d = 2$ and $d \in \{8, 32\}$	100	4096	20 000	5	ancestral or Heun, per model
Vector, misspecification	150	1024	20 000	5	as above
Fields, 16×16	300	1024	20 000	5	as above
HRRR, 32×32	300	1024	7 000 and 6 921	3	as above

Table B.4 Training and generation budget, identical for every model within a group. The optimiser is AdamW throughout, at each model specification’s own learning rate, with DLPM overriding both learning rate and step count according to its dimension regime. Sampling budgets are matched by network evaluations (NFE). Referenced from Section 6.3.

B.2 Tail-index decomposition

B.3 Moment ratios on HRRR

B.4 Configurations swept

Each configurations (except DLPM) are swept at $\text{NFE} = \{35, 50, 100, 250, 1000\}$.

Target	Working model	α_{ref}	α_{pool}	predicted	α_{gen}	implied
Field, $d = 256$	$\nu = 1.5, \beta = 0$	1.452	0.705	0.726	1.426	1.410
Misspec., $d = 32$	$\nu = 5, \beta = 0$	2.260	1.209	1.207	1.761	2.418
Misspec., $d = 32$	$\nu = 5, \beta = 1$	2.260	2.147	2.253	2.308	2.147
Skewed, $d = 32$	$\nu = 2.5, \beta = 0$	1.460	0.728	0.730	0.846	1.456
Skewed, $d = 32$	$\nu = 2.5, \beta = 0.5$	1.460	0.871	1.293	0.980	0.871
Skewed, $d = 8$	$\nu = 2.5, \beta = 0$	1.489	1.450	1.196	1.283	2.900

Table B.5 The lift reproduces its predicted latent on the misspecification and field targets; the discrepancy is in the second step and is largest at low dimension. Referenced from Section 6.5.

Target, working model	Index of $\ y\ $, by quintile of $\log \nu$				
	1	2	3	4	5
Skewed $d = 32, \nu = 1.5$	10.63	8.21	5.26	2.94	1.52
Skewed $d = 32, \nu = 2.5$	5.27	3.72	2.43	1.90	1.43
Misspec. $d = 32, \nu = 5$	28.58	18.17	16.28	13.65	6.02
Misspec. $d = 8, \nu = 5, \beta = 1$	11.51	9.72	9.63	9.27	12.69
Field $d = 256, \nu = 1.5$	≥ 50	≥ 50	≥ 50	≥ 50	≥ 50

Table B.6 Tail index of the standardised residual $\|y\| = \|x - m_\nu\|/\sqrt{\nu}$ within quintiles of the latent, all rows with $\beta_{\text{work}} = 0$ except where noted. The value 50 is the estimator’s cap, so the field rows state only that no power-law tail is detectable. Referenced from Section 6.5.

Model	w20, $\alpha = 3.53$		VIL, $\alpha = 1.14$	
	KR	SR	KR	SR
Posterior lift	0.418	0.176	0.455	0.206
t-EDM	0.280	0.401	0.532	0.304
t-Flow	1.591	0.500	0.946	0.572
DDPM	0.294	0.298	1.528	0.577
EDM	0.722	1.280	0.663	0.384
Gaussian flow	0.118	0.118	0.663	0.455
Train-against-test floor	0.376		1.028	

Table B.7 Kurtosis and skewness ratios, reported for comparability with Pandey et al. (2024, Tables 2–3) and entering no composite. By Proposition 33 the kurtosis ratio is not estimable on either channel, since $\alpha < 4$ throughout, and on VIL neither is: the floor of 1.028 is two independent samples of the same variable disagreeing by more than one hundred per cent. Referenced from Section 6.7.

Model	Parameter swept	Values
DDPM	none	—
EDM	none	—
Gaussian flow	none	—
t-EDM	degrees of freedom	target-dependent, three values
t-Flow	degrees of freedom	target-dependent, three values
DLPM	stability index	target-dependent, three values
Posterior lift	working degrees of freedom	three values
	working skewness	where the target is skewed
	latent estimator	pool, peaks-over-threshold

Table B.8 Hyperparameter grids. Grid sizes differ by design; what is equalised is the protocol, every grid being swept over the same seeds with the winner chosen by the same rule on the second reference panel. The per-target values are recorded in the run files. Referenced from Table 6.3.